北京航空航天大學 BEIHANG UNIVERSITY

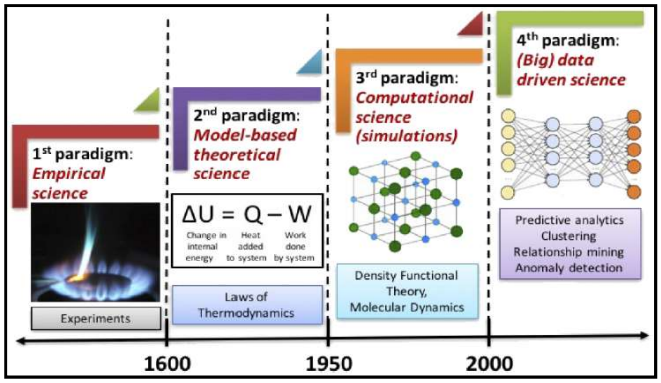
High-Throughput Computation & Big-Data Techniques in Materials Discovery
(高通量计算及大数据 vs 材料设计)

Naihua MIAO (缪奶华)

Beihang University, Beijing, China

2025-4-18

材料研究范式的演变



◆ Agrawal A, Choudhary A. Perspective: Materials informatics and big data: Realization of the "fourth paradigm" of science in materials science. APL Materials, 2016, 4(5): 053208.

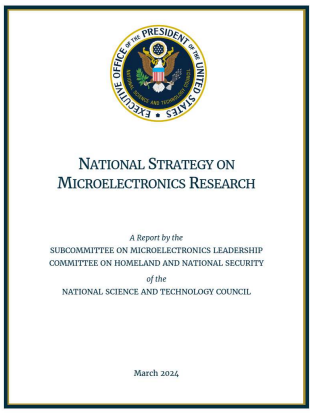
3

Outline

- ➢ **Introduction: 2D electronics/phenomena**
- ☐ **High-throughput computation in 2D materials**
- ☐ **New materials from big-data & machine learning**
- ☐ **Summary**

Introduction: 2D Electronics

4



NATIONAL STRATEGY ON MICROELECTRONICS RESEARCH

A Report by the
SUBCOMMITTEE ON MICROELECTRONICS LEADERSHIP
COMMITTEE ON HOMELAND AND NATIONAL SECURITY
of the
NATIONAL SCIENCE AND TECHNOLOGY COUNCIL

March 2024



NATIONAL STRATEGY ON MICROELECTRONICS RESEARCH

Microelectronics have become essential for many aspects of everyday life.

Semiconductors are critical to U.S. economic and national security and have become essential to many aspects of everyday life. Examples depicted here include automotive, health care, aerospace, virtual reality, financial systems, e-commerce, space satellites, defense, energy, computing, agriculture, and telecommunications. As microelectronic devices have become pervasive, their key performance requirements have become increasingly diverse, necessitating a divergence from the traditional scaling in feature size exemplified by Moore's Law. For example, the requirements for satellite applications include proven technologies that are radiation hardened, supercomputers maximize performance and speed, but devices on the edge such as sensors may prioritize low power. These application-specific requirements are driving an increased diversification of microelectronics, which will be enabled and advanced by approaches such as heterogeneous integration and chiplets.

Image credits: Adobe istock.

Introduction: 2D Electronics

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ENIAC on a chip

To illustrate the significant changes in size and scale of computing technologies, students designed and built "ENIAC-on-a-chip" to celebrate the 50th anniversary of the first programmable, electronic, general-purpose digital computer called the Electronic Numerical Integrator and Computer (ENIAC).[11] ENIAC was designed under a U.S. Army R&D program and was completed in 1945. ENIAC contained over 18,000 vacuum tubes and was approximately 8 feet tall, 3 feet deep, 100 feet long, and weighed more than 30 tons. ENIAC was programmed by hand using cables and switches, as illustrated in the figure on the left. The image on the right depicts a chip that recreated ENIAC using 0.5 µm complementary metal-oxide-semiconductor (CMOS) technology in 1995, replacing the vacuum tubes with transistors. Using today's technology, the chip would be approximately 1,000 times smaller. With respect to performance, ENIAC could do about 500 floating-point operations per second (FLOPS) while the Frontier supercomputer at Oak Ridge National Laboratory can now do more than one quintillion (1,000,000,000,000,000,000) FLOPS.

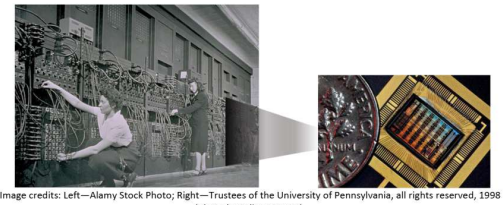


Image credits: Left—Alamy Stock Photo; Right—Trustees of the University of Pennsylvania, all rights reserved, 1998 (photo by Felice Macera).

Introduction: 2D Electronics

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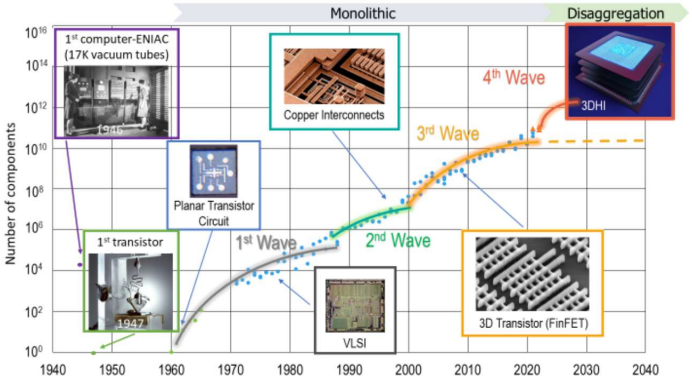
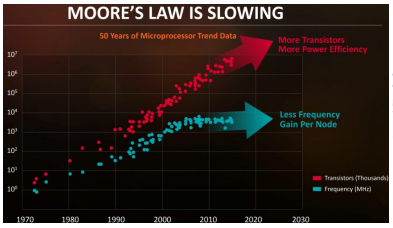


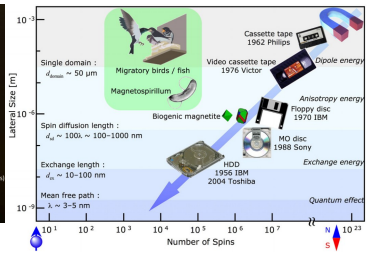
Image credit: Defense Advanced Research Projects Agency (DARPA).

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Introduction: 2D Electronics

The limit of transitional electronic and magnetic devices





IEDM 2017: AMD's grand vision

I. Zutic *et al.*, ***Rev. Mod. Phys.*** (2004) 76 323.
A. Hirohata *et al.*, ***J. Phys. D: Appl. Phys.*** 47 (2014) 193001.

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Introduction: 2D Electronics

Just how small is a transistor?

The following images show the size of transistors compared to an ant. The diameter of the circle around the picture of the ant depicts 2 millimeters (mm), or 0.002 meters. The next image is a picture taken in a scanning electron microscope (SEM) of an ant's eye, about 150 micrometers (µm), or 0.00015 meters in diameter. The third image is a picture of a hair on the ant's eye. The circle is 20 µm across. The fourth picture is a close-up SEM image of the hair, showing grooves on the hair. The diameter is 1 µm (or one thousand nanometers [nm]). A cross-section of integrated circuit transistors is shown in the top electron microscope image, illustrating modern integrated circuit transistors fit into a groove of the hair on an ant's eye. The diameter of this image is just 50 nm.

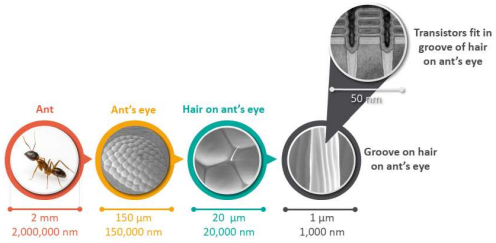


Image credits: Bottom images—National Institute of Standards and Technology (NIST); Upper right image—reprint courtesy of IBM Corporation ©.

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Introduction: 2D Electronics

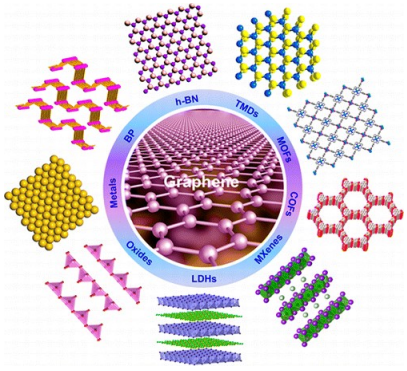
Coordination among entities in the semiconductor materials and associated research ecosystem, including national labs, private sector companies, and consortia, will provide pathways to deployment of advanced materials for devices, interconnects, circuits, and systems. Frameworks like the Materials Innovation Infrastructure developed as part of the Materials Genome Initiative (MGI)" can play an important role in organizing the materials community around grand challenges in developing new capabilities or functional enhancements for microelectronics.

Elements of advanced materials R&D needed to support new capabilities include:

- Focused research on emerging organic and inorganic materials including two-dimensional (2D) materials exhibiting quantum effects/properties; wide-bandgap and ultra-wide-bandgap materials for energy-efficient electronics and use in extreme environments; materials optimized for high-bandwidth interconnects; materials for ultra-high-frequency operation (optical, electrical, and electromechanical); materials that enable non-von Neumann architectures; and biotic-abiotic hybrid systems.
- Exploration of materials that can be seamlessly integrated into existing process flows, and that can, for example, add functionality into the back-end-of-line (BEOL) to improve performance and allow for greater 3D integration.
- Increased efforts to improve existing bulk substrate materials and to accelerate the development and deployment of new ones.
- Unified semiconductor materials data infrastructure to facilitate knowledge sharing and accelerate innovation.
- New modeling, characterizatior accurate determination of *all* of
- R&D efforts to accelerate the production-worthy tools for new
- New measurements and standa accelerate materials research an
- High-throughput experimentation and modelling methods, coupled with new capabilities in optical, electron, and scanning probe microscopy inspection tools to improve speed, throughput, yield, precision, and accuracy.
- Hybrid metrology methods that combine data from multiple measurement tools integrated with new ML methods to utilize the data and enable process optimization.
- Integrated AI/ML/physics-based models capable of digesting a fab's worth of real-time process data for advanced predictive analytics to measure and improve yield and enable fab virtualization in semiconductor and microelectronics manufacturing.

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Introduction: 2D Materials



Many exotic features:

- ✓Thickness;
- ✓Surface area;
- ✓Quantum effects;
- ✓…

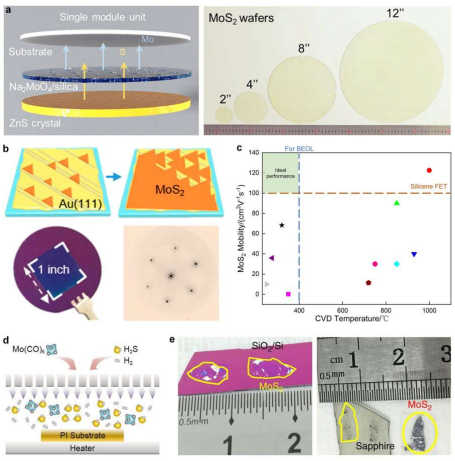
But:

- ➢Band gap;
- ➢Magnetism;
- ➢Stability;
- ➢…

K. S. Novoselov *et al*., ***Science***, 2004, 306, 666;
J. Qiao *et al*, ***Nature Commun***., 2014, 5, 4475; **H. Zhang**, ***ACS Nano***, 2015, *9* (10), pp 9451–9469.

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Introduction: 2D Materials

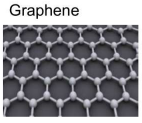


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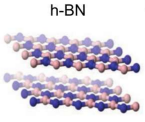
Introduction: 2D Materials

Layered materials have a vast potential in applications

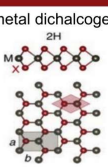
Graphene

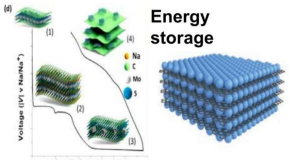


h-BN



Transition metal dichalcogenides (TMDs)

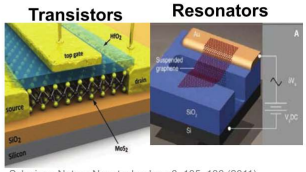




Kumar et al. ECS J. Solid State Sci. Technol. 2016
Raccichini et al, Nature Materials (2015)

Transistors

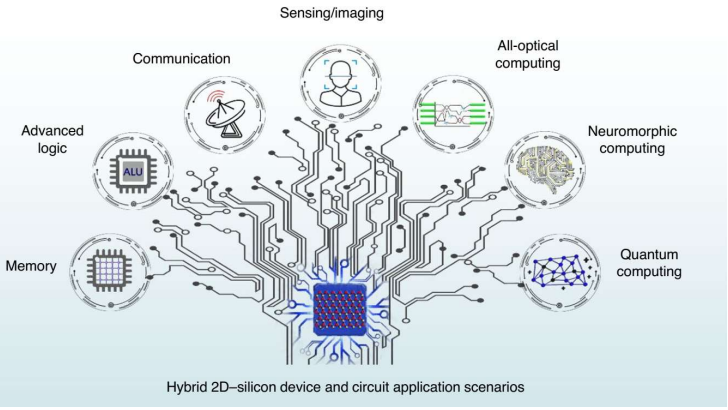
Resonators



Schwierz, Nature Nanotechnology 6, 135–136 (2011)
Bunch et al, Science 315, 490 (2007)

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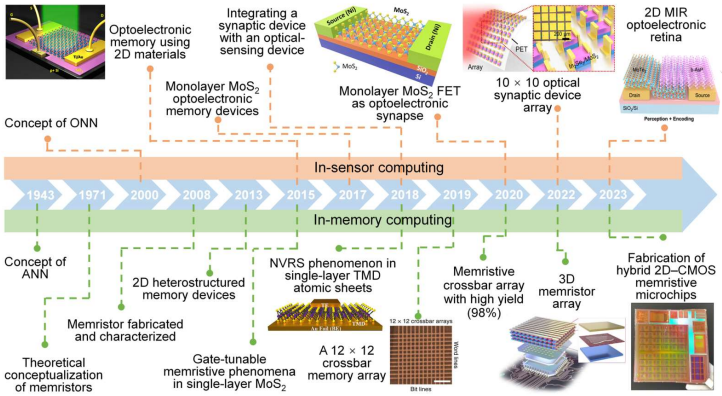
Introduction: 2D Electronics



DOI: 10.1038/s41563-022-01383-2

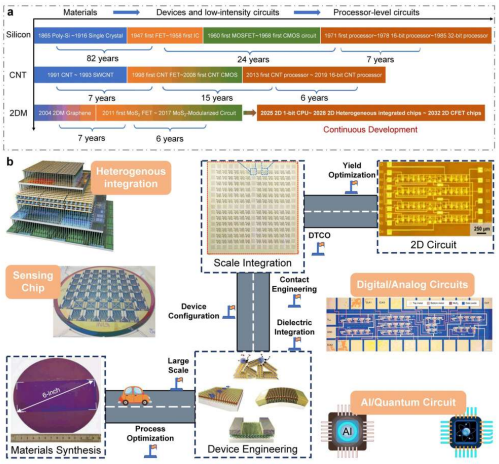
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Introduction: 2D Electronics



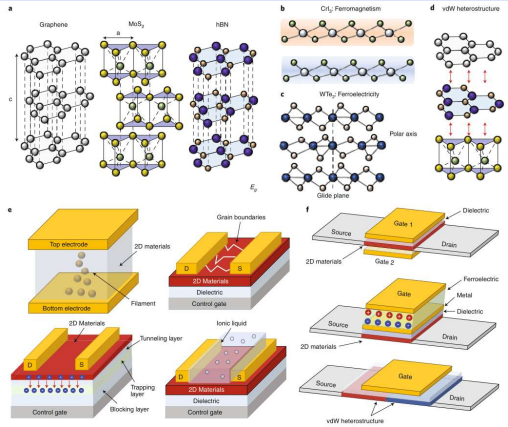
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Introduction: 2D Electronics



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Introduction: 2D Materials for Next-Generation Computing Technologies



DOI: 10.1038/s41565-020-0724-3

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Introduction: 2D Materials for Next-Generation Computing Technologies

nature

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Article | Published: 02 April 2025

A RISC-V 32-bit microprocessor based on two-dimensional semiconductors

Mingrui Ao, Xiucheng Zhou, Xinjie Kong, Saifei Gou, Sifan Chen, Xiangqi Dong, Yuxuan Zhu, Qicheng Sun, Zhejia Zhang, Jinshu Zhang, Qiran Zhang, Yan Hu, Chuming Sheng, Kaixuan Wang, Shuiyuan Wang, Jing Wan, Jun Han, Wenzhong Bao ✉ & Peng Zhou ✉

Nature **640**, 654–661 (2025) | Cite this article

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Introduction: 2D Materials for Next-Generation Computing Technologies

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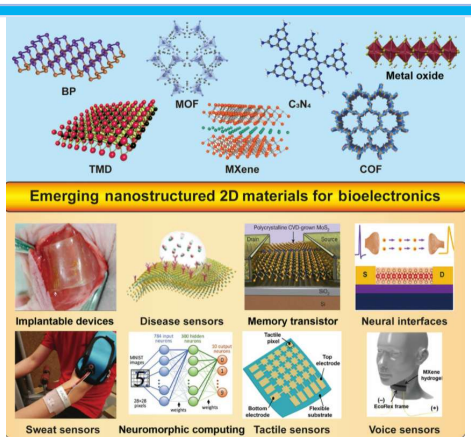
Subnanosecond flash memory enabled by 2D-enhanced hot-carrier injection

Yutong Xiang, Chong Wang, Chunsen Liu ✉, Tanjun Wang, Yongbo Jiang, Yang Wang, Shuiyuan Wang & Peng Zhou ✉

Nature (2025) | Cite this article

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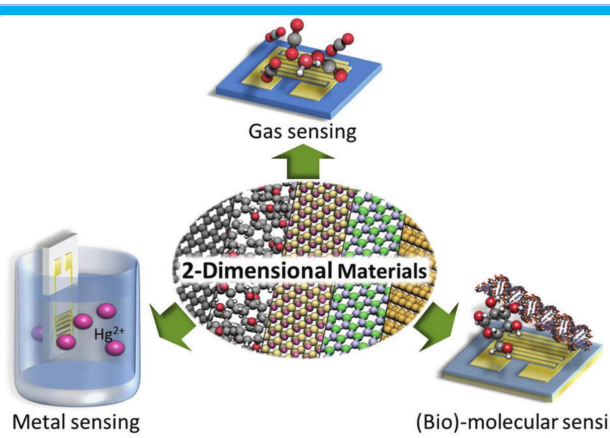
Introduction: 2D Materials for Bioelectronics



DOI:10.1002/inf2.12299

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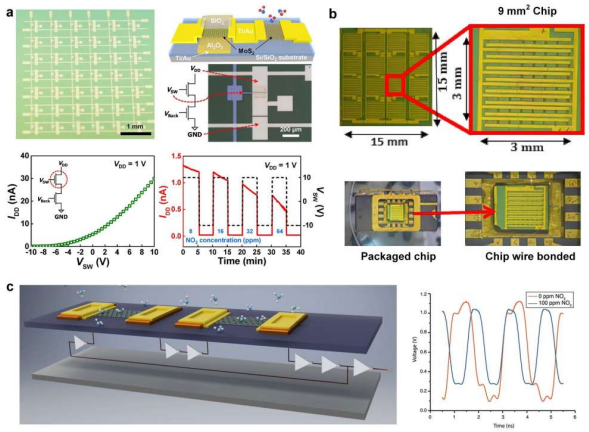
Introduction: 2D Materials for Sensing



DOI:10.1039/c8cs00417j

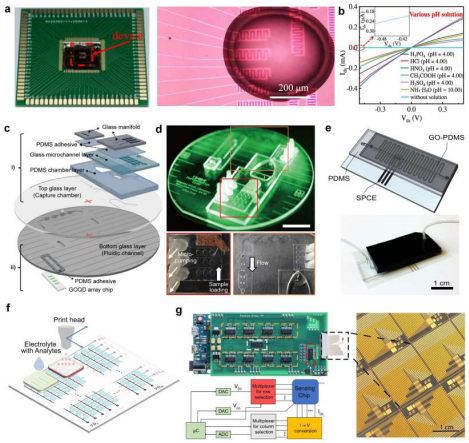
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Introduction: 2D Materials for Sensing



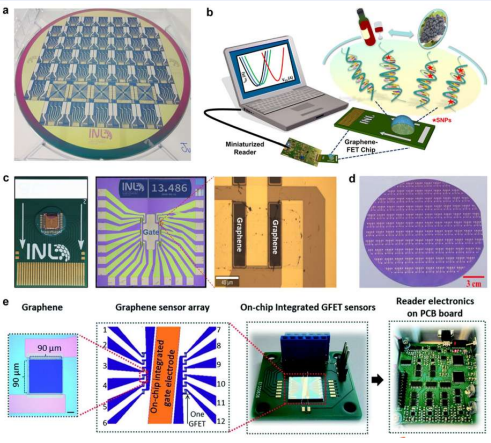
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Introduction: 2D Materials for Sensing



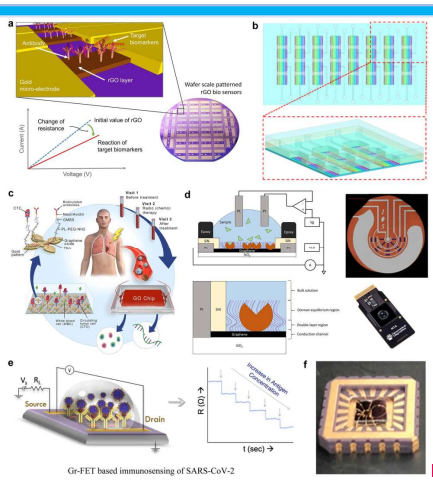
23

Introduction: 2D Materials for Sensing



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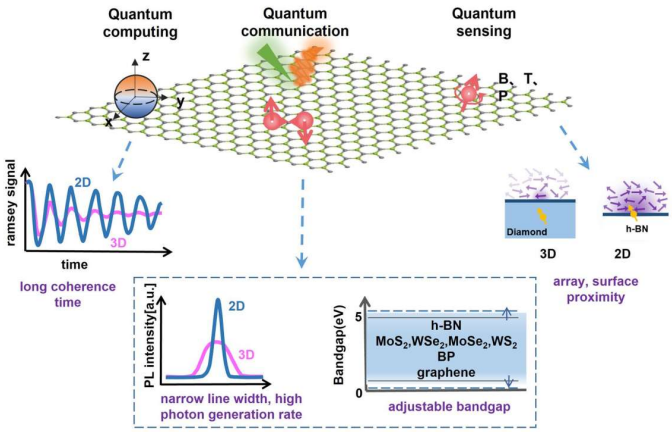
Introduction: 2D Materials for Sensing



Gr-FET based immunosensing of SARS-CoV-2

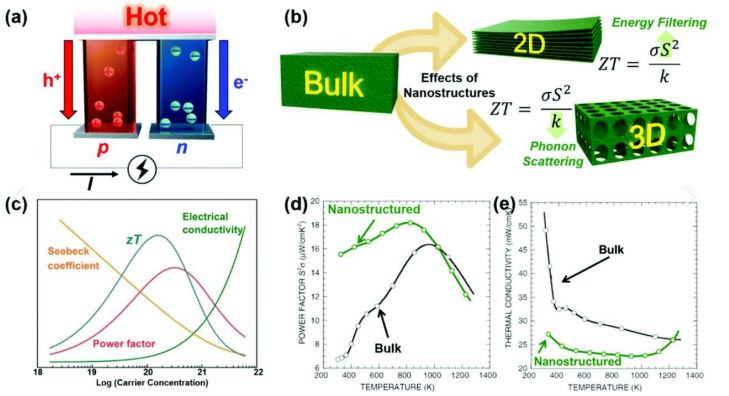
25

Introduction: 2D Materials for Quantum…



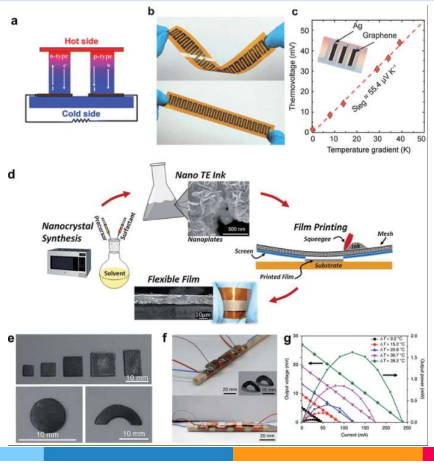
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Introduction: 2D Materials for Thermoelectrics



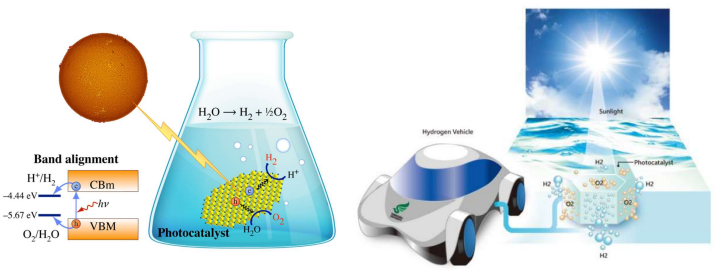
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Introduction: 2D Materials for Thermoelectrics



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Introduction: 2D Materials for Water Splitting



J. Phys. Chem. Lett. 2015, 6, 1087−1098

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Outline

- Introduction: 2D spintronics/phenomena
- ➢ High-throughput computation in 2D materials
- New 2D materials from big-data & machine learning
- Summary

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研究案例①：二维材料高通量剥离

ARTICLES
https://doi.org/10.1038/s41565-017-0035-5
nature nanotechnology

Two-dimensional materials from high-throughput computational exfoliation of experimentally known compounds

Nicolas Mounet[1]*, Marco Gibertini[1], Philippe Schwaller[1], Davide Campi[1], Andrius Merkys[1,2], Antimo Marrazzo[1], Thibault Sohier[1], Ivano Eligio Castelli[1], Andrea Cepellotti[1], Giovanni Pizzi[1] and Nicola Marzari[1]*

Two-dimensional (2D) materials have emerged as promising candidates for next-generation electronic and optoelectronic applications. Yet, only a few dozen 2D materials have been successfully synthesized or exfoliated. Here, we search for 2D materials that can be easily exfoliated from their parent compounds. Starting from 108,423 unique, experimentally known 3D compounds, we identify a subset of 5,619 compounds that appear layered according to robust geometric and bonding criteria. High-throughput calculations using van der Waals density functional theory, validated against experimental structural data and calculated random phase approximation binding energies, further allowed the identification of 1,825 compounds that are either easily or potentially exfoliable. In particular, the subset of 1,036 easily exfoliable cases provides novel structural prototypes and simple ternary compounds as well as a large portfolio of materials to search from for optimal properties. For a subset of 258 compounds, we explore vibrational, electronic, magnetic and topological properties, identifying 56 ferromagnetic and antiferromagnetic systems, including half-metals and half-semiconductors.

NATURE NANOTECHNOLOGY

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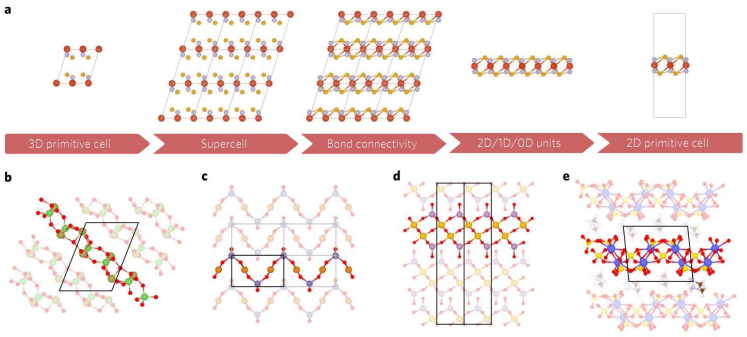
Table 1 | Database statistics

	Unique to the ICSD	Unique to the COD	Common to both	Total sum
Experimental data				
CIF inputs	99,212	87,070		186,282
Unique 3D structures (set A)	34,548	60,354	13,521	108,423
Layered 3D structures (set B)	3,257	1,180	1,182	5,619
DFT calculations				
Layered 3D, relaxed (set C)	2,165	175	870	3,210
Binding energies (set D)	1,795	126	741	2,662
2D easily exfoliable (EE)	663	79	294	1,036
2D potentially exfoliable (PE)	524	34	231	789
Total	1,187	113	525	1,825

Experimental data: number of structures imported from the two databases (ICSD or COD, uniqueness not tested), number of unique 3D structures in each imported set or common to both (set A), and number of layered 3D structures identified using the geometrical criteria discussed in the text (set B). DFT calculations: number of structures that were relaxed (set C), number of structures that remain classified as layered after relaxation and for which binding energies were computed (set D), and number of easily or potentially (see text) exfoliable compounds.

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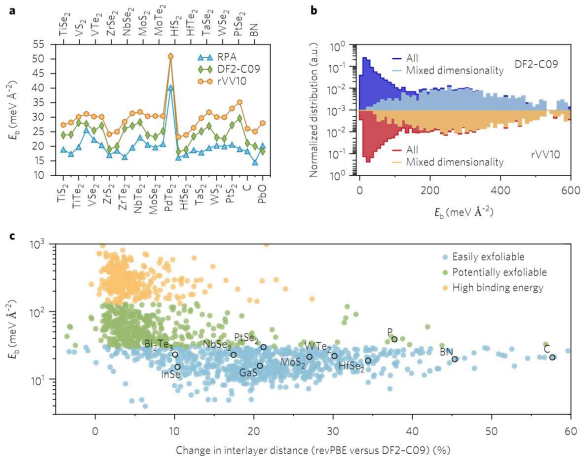
Structures and procedure



Screening for low-dimensional manifolds in a parent 3D crystal

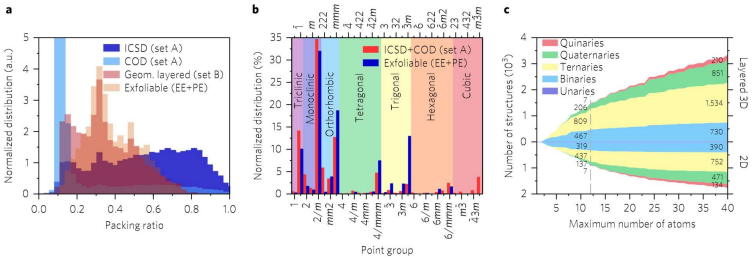
34

Binding energies of bulk compounds



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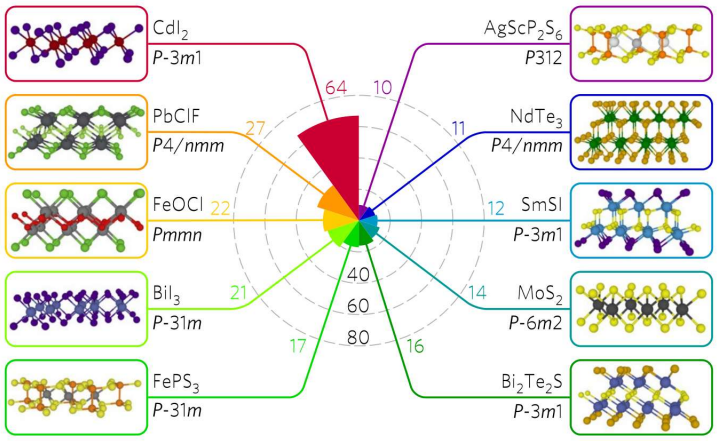
Statistics on the 2D and 3D databases



a, Packing ratios for the 3D structures considered according to their ICSD or COD provenance, layered compounds (set B) and exfoliable materials (EE+ PE).
b, Distribution of point groups in the set of structures exported from the ICSD and COD (set A) and in the set of easily and potentially exfoliable layered materials (EE+ PE). The different colours mark the different crystal systems.
c, Number of structures as a function of the number of atoms in the primitive cell (top, layered 3D structures; bottom, 2D structures in the EE+ PE groups).

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Most common 2D structural prototypes



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Exfoliable ferromagnetic compounds

Table 2 | Easily exfoliable magnetic compounds

	Ferromagnetic	Antiferromagnetic
Metals	$Co(OH)_2$, CoO_2, ErHCl, ErSeI, EuOBr, EuOI, $FeBr_2$, FeI_2, FeTe, LaCl, NdOBr, PrOBr, ScCl, SmOBr, SmSI, TbBr, TmI_2, TmOI, VS_2, VSe_2, VTe_2, YCl, YbOBr, YbOCl	CoI_2, $CrSe_2$, FeO_2, FeOCl, FeSe, PrOI, VOBr
Semiconductors	CdOCl, $CoBr_2$, $CoCl_2$, CrOBr, CrOCl, CrSBr, $CuCl_2$, ErSCl, HoSI, $LaBr_2$, $NiBr_2$, $NiCl_2$, NiI_2	$CrBr_2$, CrI_2, LaBr, $Mn(OH)_2$, $MnBr_2$, $MnCl_2$, MnI_2, VBr_2, VCl_2, VI_2, $VOBr_2$, $VOCl_2$

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Supporting information: 275 pages

Two-dimensional materials from high-throughput computational exfoliation of experimentally known compounds

Nicolas Mounet, Marco Gibertini, Philippe Schwaller, Davide Campi, Andrius Merkys, Antimo Marrazzo, Thibault Sohier, Ivano Eligio Castelli, Andrea Cepellotti, Giovanni Pizzi, and Nicola Marzari

Supplementary Summary

Definitions and further details	3
Supplementary Figures	4
Supplementary Tables	6
Structures	16
Supplementary References	274

结构

声子谱

电子能带

磁距等信息

Nature Nanotechnology **13**, 246–252(2018)
https://doi.org/10.1038/s41565-017-0035-5

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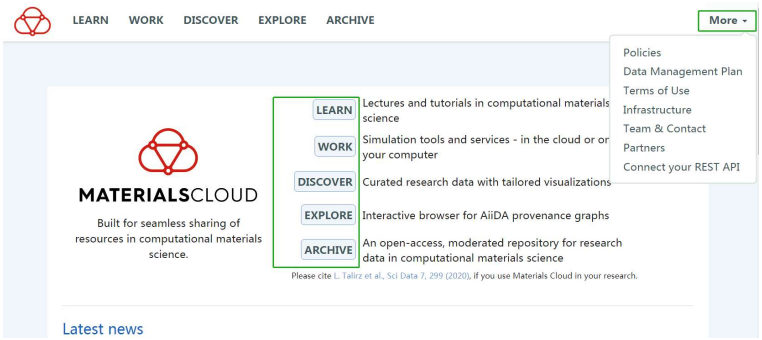
2D Materials Databases



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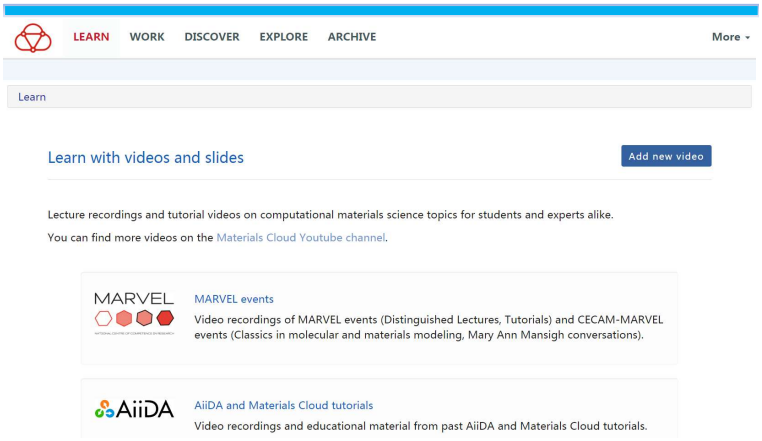
2D Materials Databases



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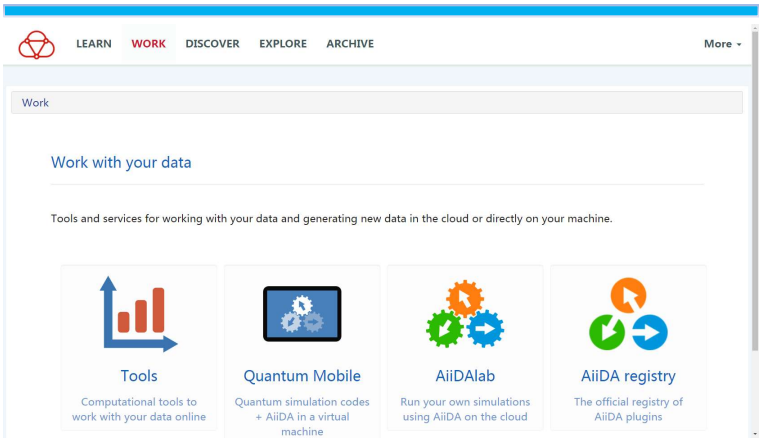
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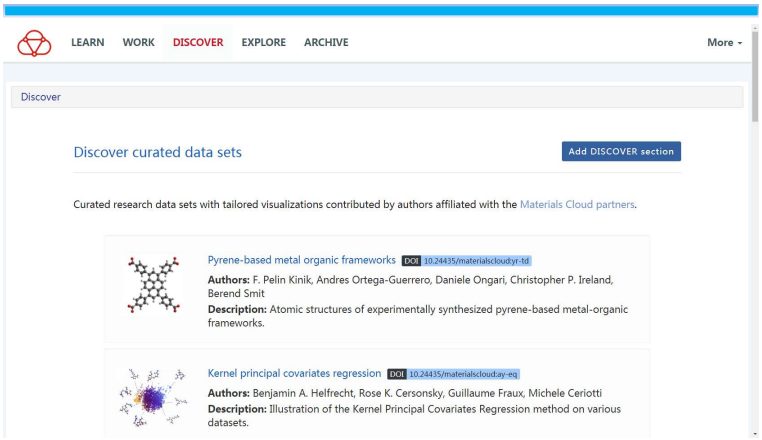
2D Materials Databases



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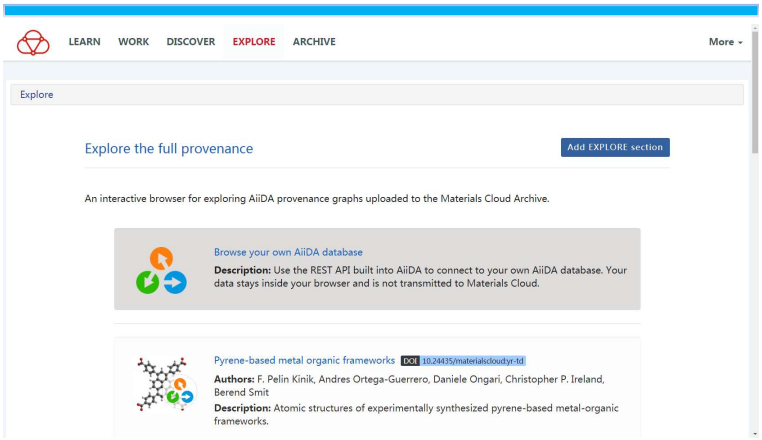
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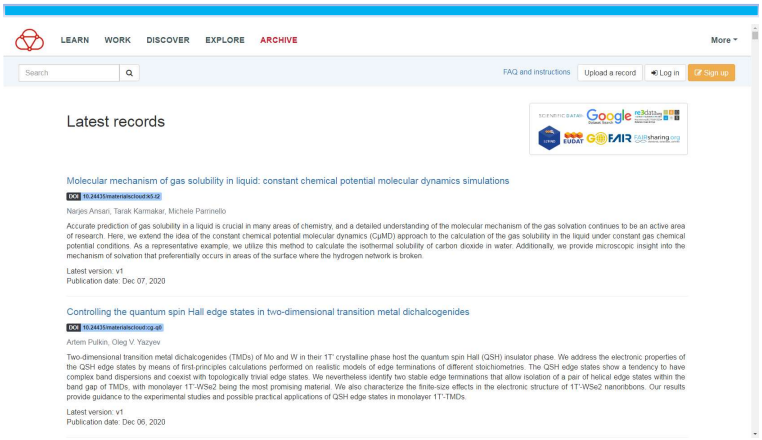
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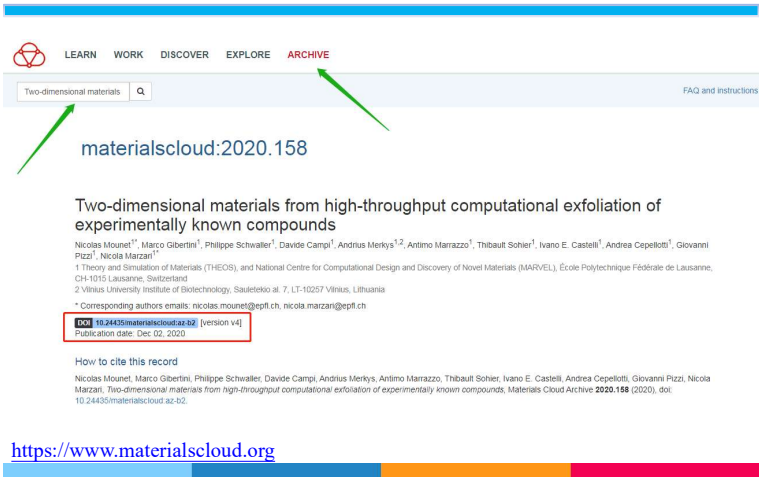
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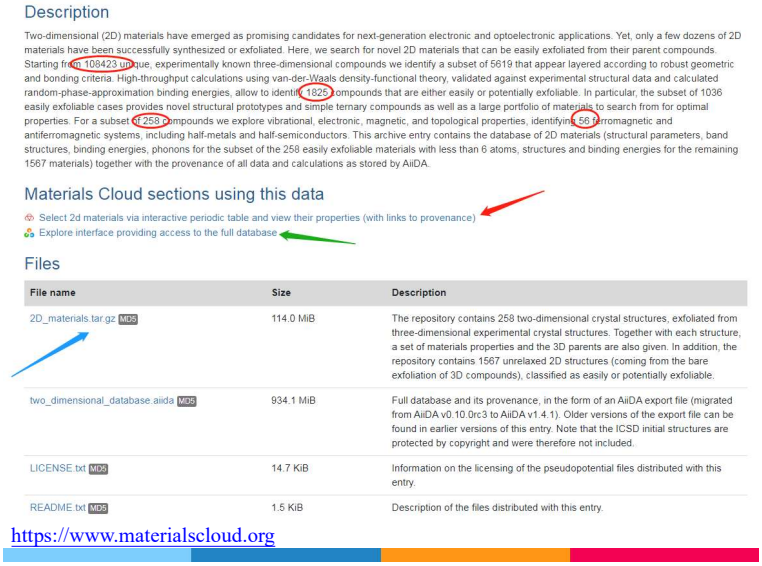
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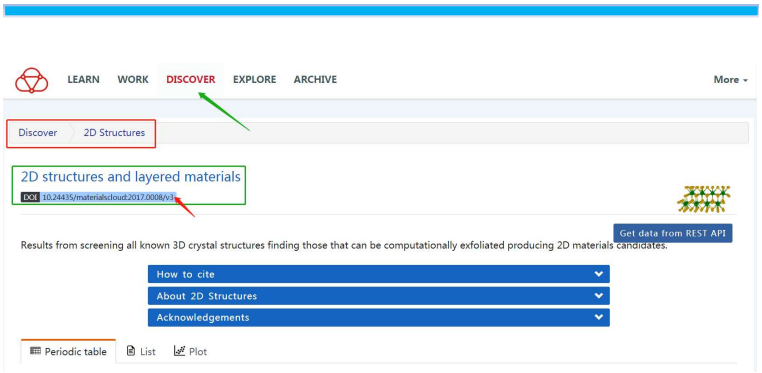
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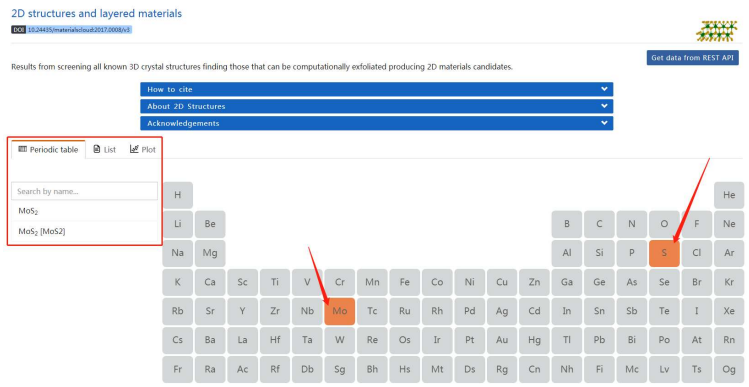
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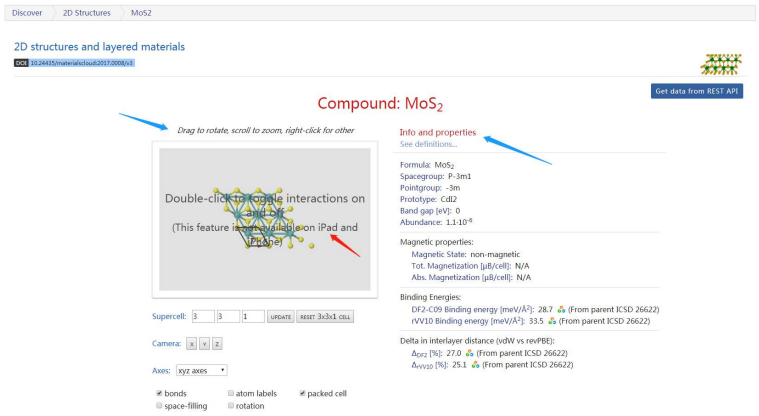
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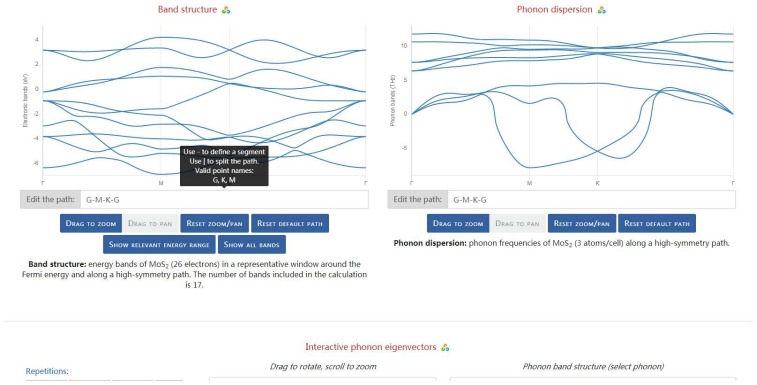
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2D Materials Databases



材料晶体结构信息：
元素种类
晶格常数
原子坐标

https://www.materialscloud.org

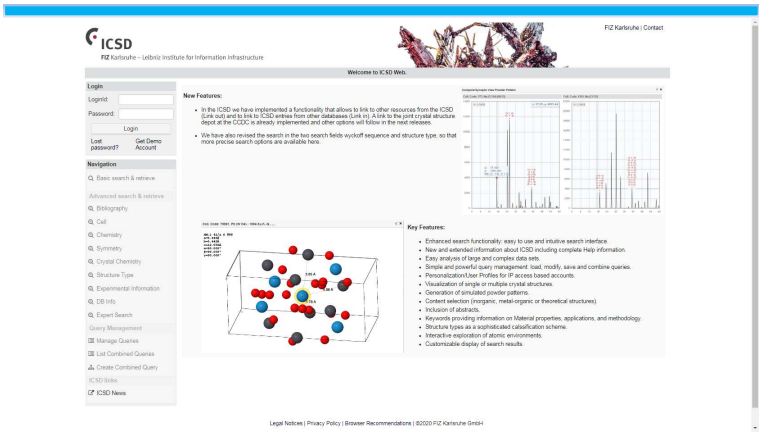
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Summary-1

- Unknown 2D materials from known 3D bulk phases database
- New 2D magnetic materials
- New 2D materials database
- Potential application of materials database:
  Data mining / machine learning… to design new materials…

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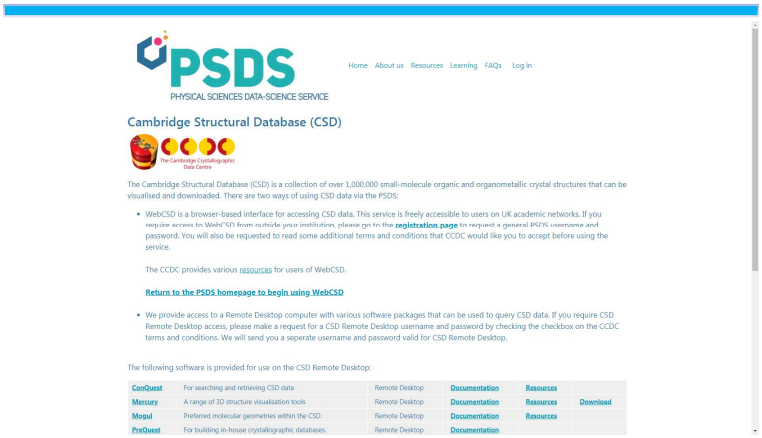
一些常用的材料数据库



http://www2.fiz-karlsruhe.de/icsd_home.html

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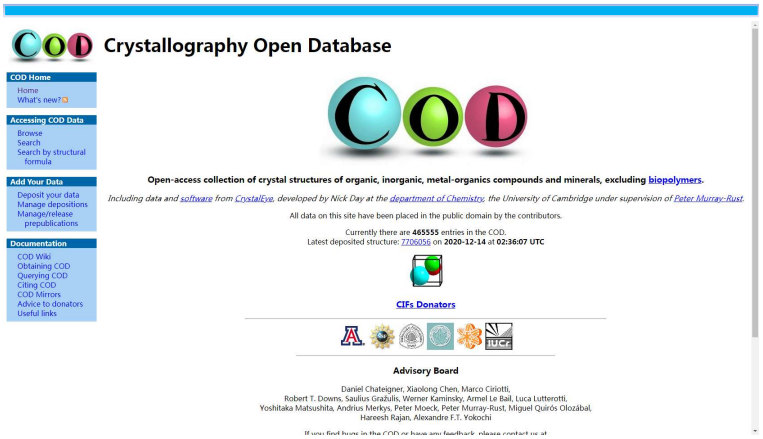
一些常用的材料数据库



https://www.psds.ac.uk/csd

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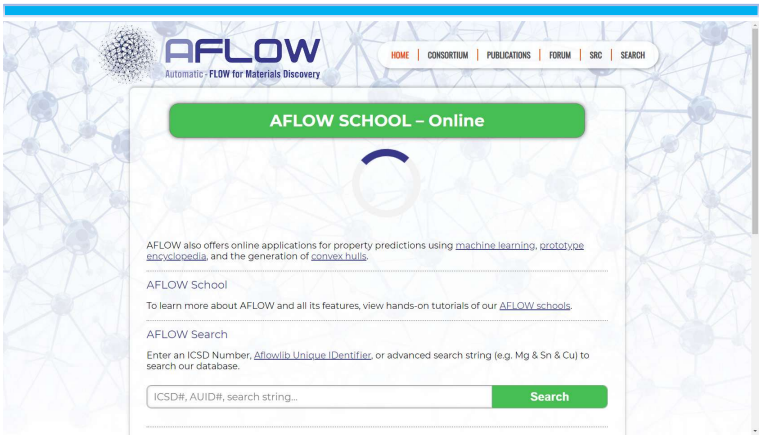
一些常用的材料数据库



http://www.crystallography.net/cod

58

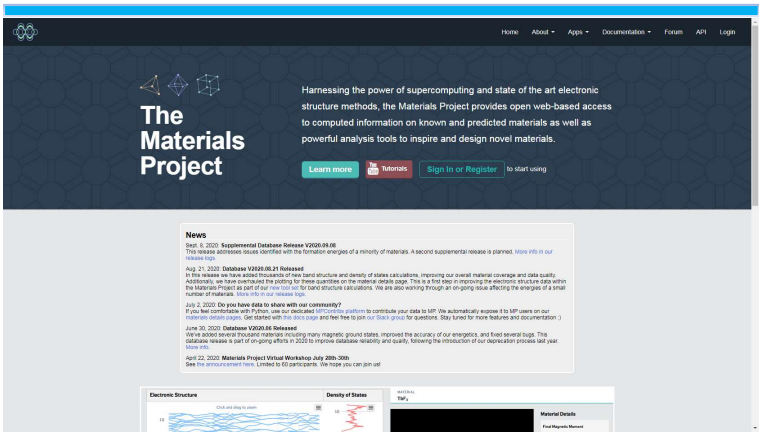
一些常用的材料数据库



http://www.aflowlib.org

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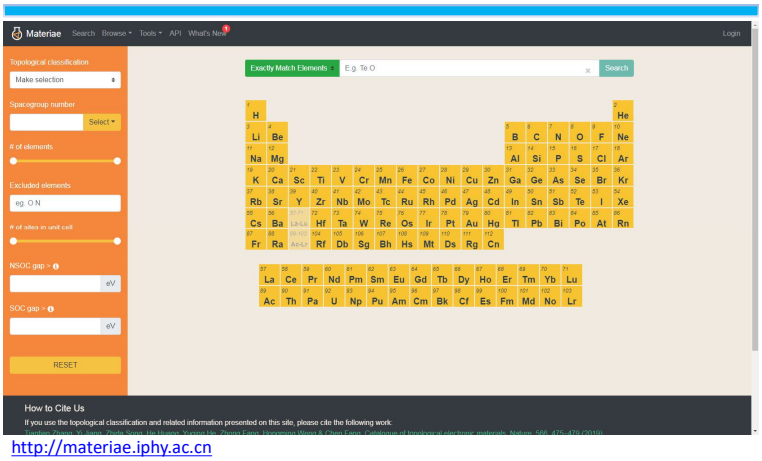
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https://www.materialsproject.org

60

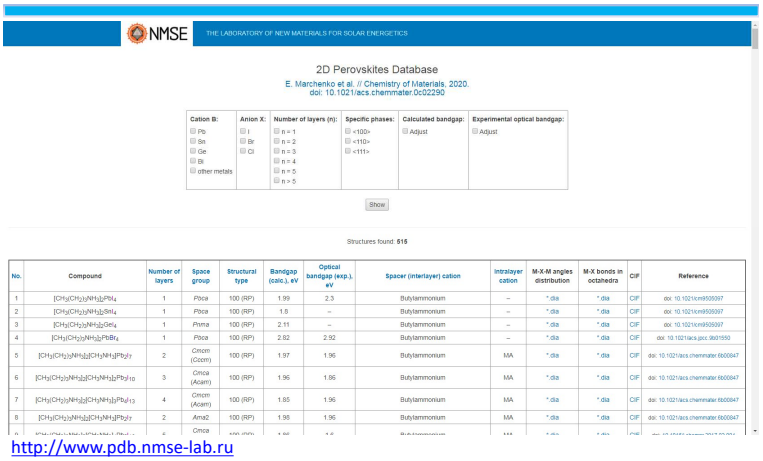
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http://materiae.iphy.ac.cn

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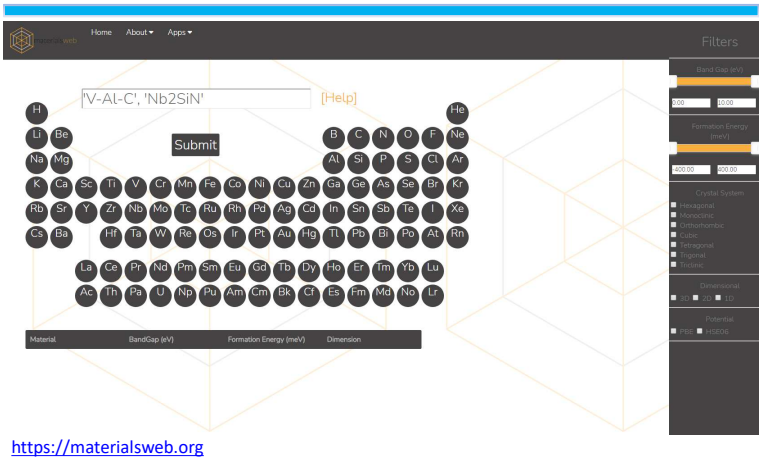
一些常用的材料数据库



http://www.pdb.nmse-lab.ru

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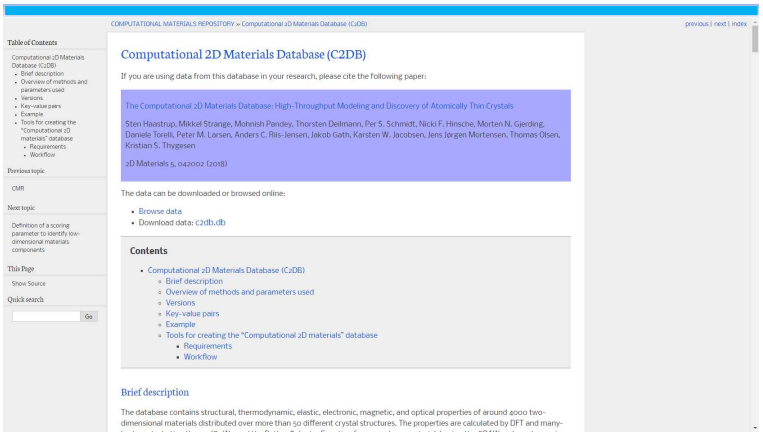
一些常用的材料数据库



https://materialsweb.org

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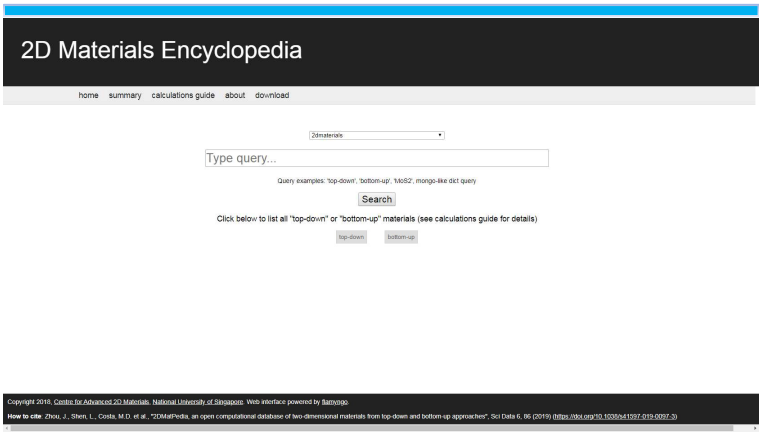
一些常用的材料数据库



https://cmr.fysik.dtu.dk/c2db/c2db.html

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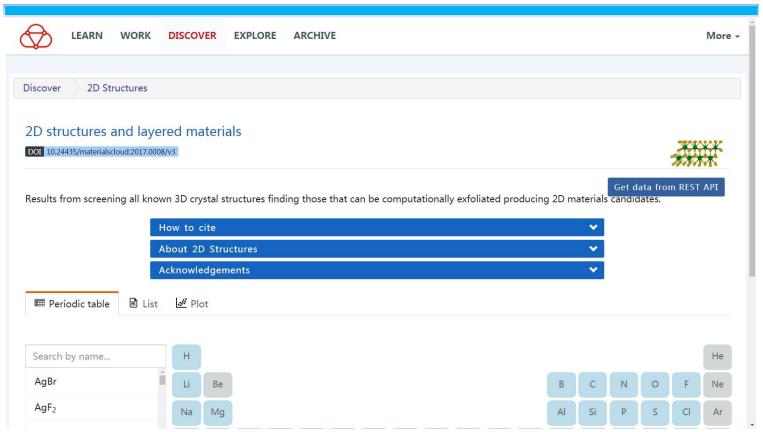
一些常用的材料数据库



http://www.2dmatpedia.org

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一些常用的材料数据库



https://www.materialscloud.org/discover/2dstructures/dashboard/ptable

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Outline

- Introduction: 2D spintronics/phenomena
- High-throughput computation in 2D materials
- ➢ New materials from big-data & machine learning
- Summary

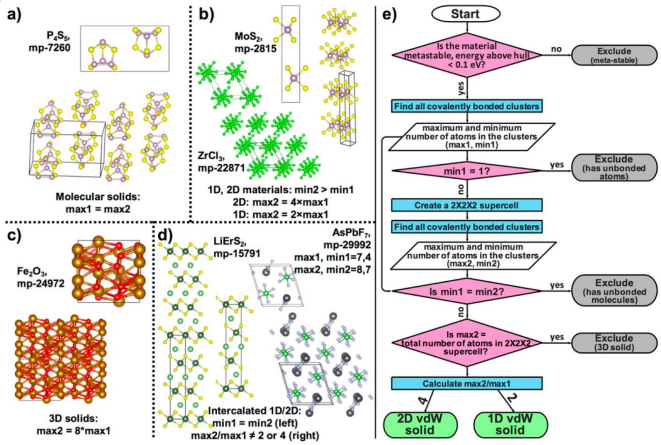
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研究案例②：弱键固体的数据挖掘

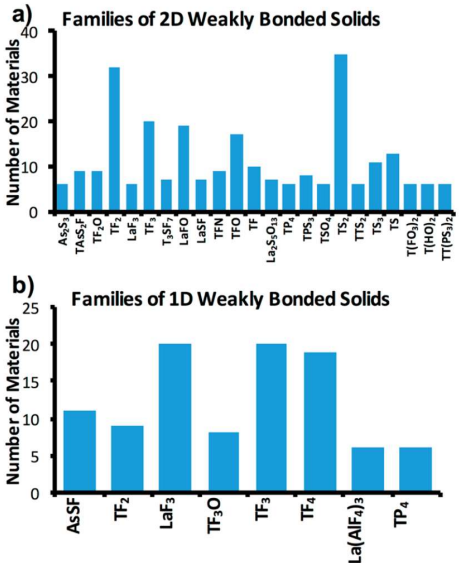


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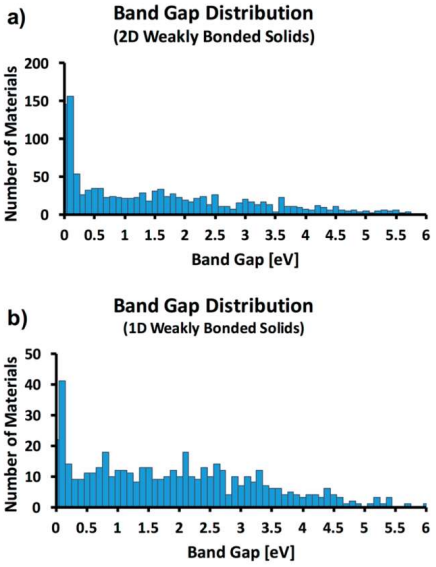
Relationship between material dimensionality and covalently bonded cluster sizes



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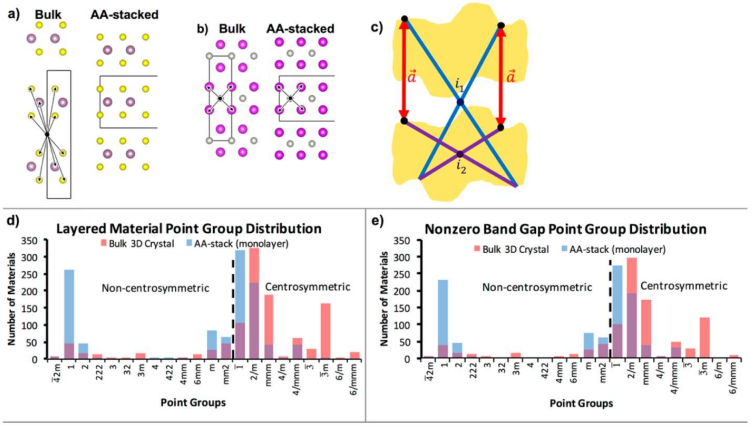


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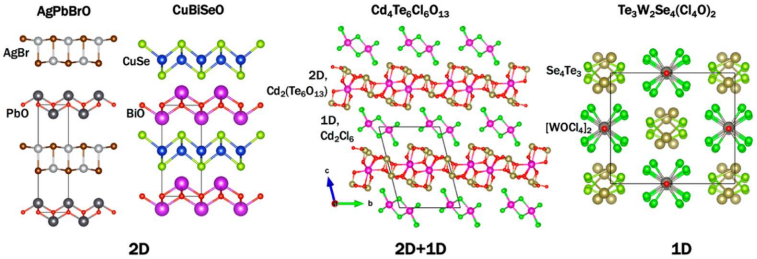
71

Clusters, inversion symmetry, statics



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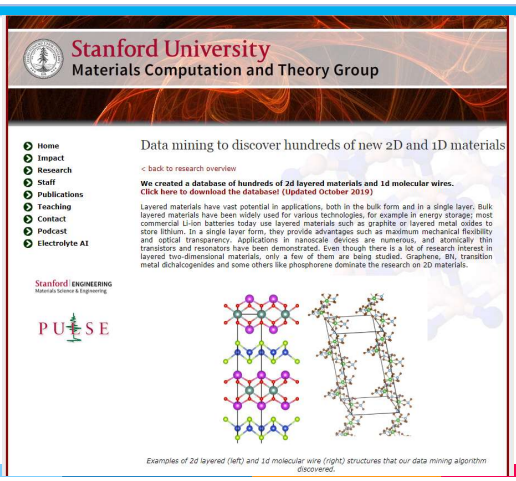
Examples of lattice-commensurate heterostructures



DOI:10.1021/acs.nanolett.6b05229
Nano Lett. 2017, 17, 1915−1923

73

Examples of lattice-commensurate heterostructures



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研究案例③:机器学习二维铁磁体

COMMUNICATION

ADVANCED MATERIALS www.advmat.de

Check for updates

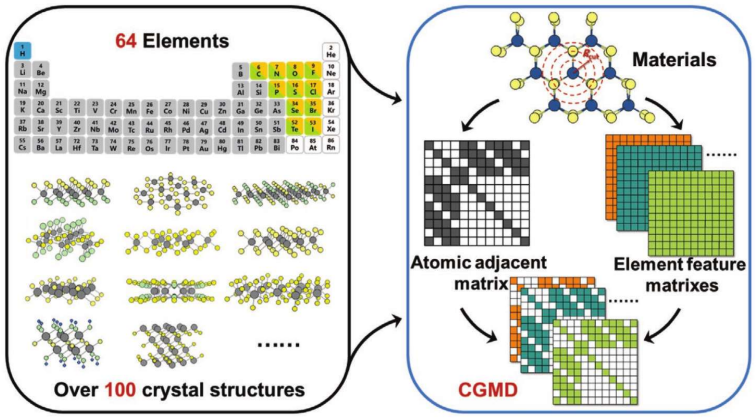
Coupling a Crystal Graph Multilayer Descriptor to Active Learning for Rapid Discovery of 2D Ferromagnetic Semiconductors/Half-Metals/Metals

*Shuaihua Lu, Qionghua Zhou, Yilv Guo, Yehui Zhang, Yilei Wu, and Jinlan Wang**

2D ferromagnetic (FM) semiconductors/half-metals/metals are the key materials toward next-generation spintronic devices. However, such materials are still rather rare and the material search space is too large to explore exhaustively. Here, an adaptive framework to accelerate the discovery of 2D intrinsic FM materials is developed, by combining advanced machine-learning (ML) techniques with high-throughput density functional theory calculations. Successfully, about 90 intrinsic FM materials with desirable bandgap and excellent thermodynamic stability are screened out and a database containing 1459 2D magnetic materials is set up. To improve the performance of ML models on small-scale datasets like diverse 2D materials, a crystal graph multilayer descriptor using the elemental property is proposed, with which ML models achieve prediction accuracy over 90% on thermodynamic stability, magnetism, and bandgap. This study not only provides dozens of compelling FM candidates for future spintronics, but also paves a feasible route for ML-based rapid screening of diverse structures and/or complex properties.

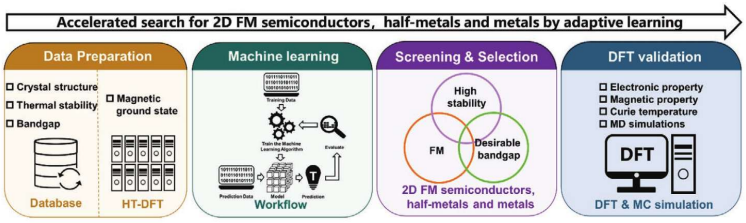
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Machine learning: 2D ferromagnetic materials



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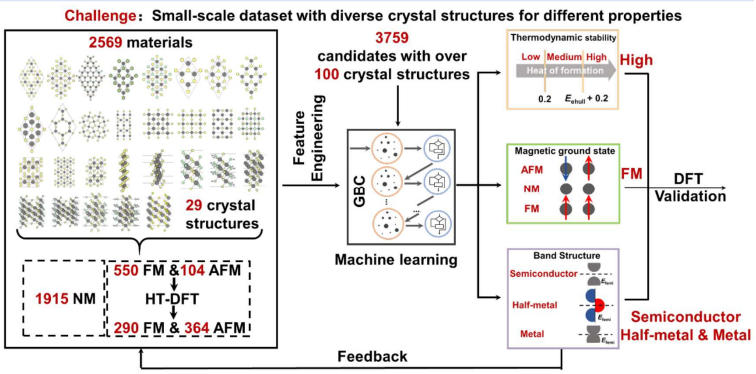
Machine learning: 2D ferromagnetic materials



Material discovery workflow with feedback loop combined with adaptive learning and DFT for searching stable 2D FM materials.

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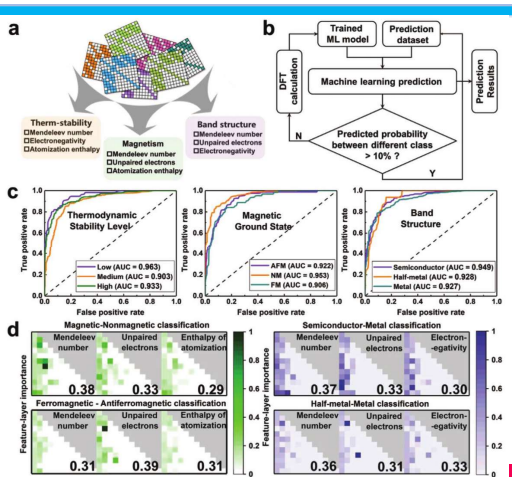
Machine learning: 2D ferromagnetic materials



Detailed material discovery workflow with feedback loop combined with ML and DFT

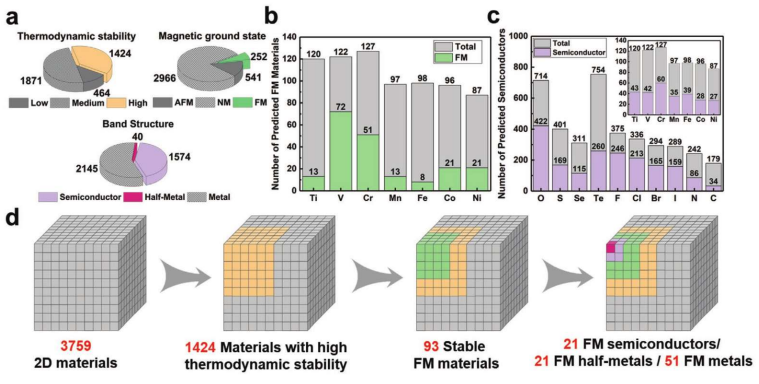
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Machine learning: 2D ferromagnetic materials



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Machine learning: 2D ferromagnetic materials



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Table 1. Magnetic and electronic properties of 20 FM semiconductors.

Formula	Space group	M_{tot} [a]	E_{FM-AFM} [a]	E_g [a]	E_{MAE} [a]	T_C [a]
$Mn(CO_3)_2$	P-3	2.9	−16.1	1.64	73.9	55
CrF_2	$P2_1/c$	7.5	−4.1	1.29	346.1	19
$CrMoCl_6$	P312	5.5	−12.0	0.96	266.4	35
$CrMoBr_6$	P312	5.6	−21.9	0.85	583.5	65
$CrIn(ClO)_2$	Pmm2	2.9	−13.7	0.77	173.3	<5
$CrMoI_6$	P312	5.5	−20.9	0.76	1267.9	70
$CrPS_4$	C2/m	5.5	−13.2	0.70	52.3	37
$V_4S_9Br_4$	P4/nmm	1.9	−51.2	0.59	488.6	<5
CrOF	Pmmn	5.8	−25.2	0.58	148.4	42
$CrSiTe_3$	P-31m	5.8	−3.5	0.57	209.9	12
$CrWCl_6$	P312	5.5	−33.7	0.63	2301.9	97
$CrWBr_6$	P312	5.5	−42.9	0.56	2590.2	144
$CrWI_6$	P312	5.4	−43.6	0.50	3535.4	148
$Co(ClO_4)_2$	P-3	2.9	−2.7	0.44	446.2	13
$CrGeTe_3$	P-31M	5.7	−20.0	0.47	138.7	52
$CrBr_2$	$P2_1/c$	7.4	−16.0	0.27	229.6	30
$VAg(PSe_3)_2$	C2	1.7	−9.8	0.23	161.9	68
VBr_4	P4/mbm	1.2	−15.2	0.23	670.2	33
CoF_4	P4/mbm	5.7	−41.5	0.21	707.8	92
Mn_2FeO_6	P-31m	3.9	−31.3	0.16	–	–

[a]The space group, the net magnetic moment (M_{tot}, μ_B/f.u.), the energy difference between FM and the lowest AFM states (E_{FM-AFM}, meV/f.u.), the bandgap value (E_g, eV), the magnetic anisotropy energies (E_{MAE}, μeV per magnetic metal) and the Curie temperature (T_C, K).

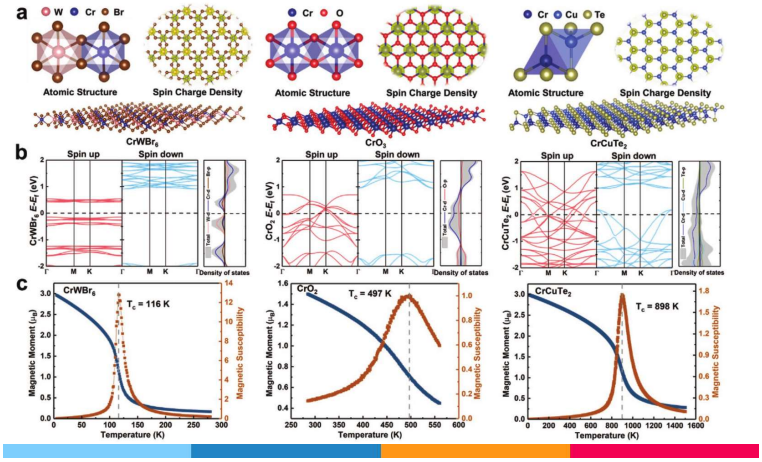
Table 2. Magnetism and electronic properties of 21 FM half-metals.

Formula	Space group	M_{tot}	E_{FM-AFM}	E_g^{spin} [a]	E_{MAE}	T_C
Mn_3F_8	P-1	12.5	−20.1	4.58	515.3	47
V_2CoO_6	C2/m	2.9	−16.1	4.58	1417.9	19
CrO_2	P-3m1	1.9	−143.6	3.24	70.4	497
V_2Cl_5	P-1	4.6	−100.0	3.09	214.6	11
V_3Br_8	P-3m1	6.6	−12.5	2.86	133.9	25
VO_2	$P2_1/m$	1.9	−141.7	2.85	80.1	356
V_2Br_5	P-1	4.6	−107.8	2.43	293.6	14
$V(CO_3)_2$	P-3	0.9	−40.6	2.27	65.2	141
V_3TeCl_7	P3m1	5.7	−59.1	2.01	1103.6	114
V_3I_8	P3m1	6.6	−57.5	1.96	977.1	120
Mn_2CrO_6	C2/m	7.7	−23.0	1.96	327.2	23
Mn_2CrO_6	P-3m1	7.8	−12.1	1.95	110.8	21
V_2I_5	P-1	4.6	−119.5	1.87	867.0	22
V_3SBr_7	P3m1	5.6	−58.0	1.87	–	–
$TcCl_3$	P-1	3.7	−24.8	1.66	5119.8	89
TiI_3	P-3m1	1.8	−22.8	1.61	1671.9	60
$CrCl_4$	P-1	3.9	−49.0	1.44	2083.7	111
$CrBr_4$	P-1	4.0	−71.5	1.28	9504.1	178
Mn_3CrO_8	P-3m1	10.7	−53.0	1.22	97.9	83
Mn_2CoO_6	C2/m	4.9	−49	1.95	–	–
Mn_3CoO_8	P-3m1	7.8	−34.9	1.22	–	–

[a]The spin bandgap (E_g^{spin}, eV).

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Machine learning: 2D ferromagnetic materials



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研究案例④:机器学习二维材料表征



Deep-Learning-Enabled Fast Optical Identification and Characterization of 2D Materials

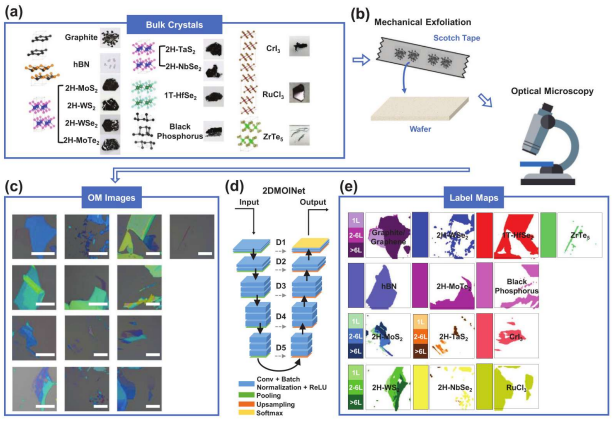
Bingnan Han, Yuxuan Lin, Yafang Yang, Nannan Mao, Wenyue Li, Haozhe Wang, Kenji Yasuda, Xirui Wang, Valla Fatemi, Lin Zhou, Joel I.-Jan Wang, Qiong Ma, Yuan Cao, Daniel Rodan-Legrain, Ya-Qing Bie, Efrén Navarro-Moratalla, Dahlia Klein, David MacNeill, Sanfeng Wu, Hikari Kitadai, Xi Ling, Pablo Jarillo-Herrero,* Jing Kong,* Jihao Yin,* and Tomás Palacios**

Advanced microscopy and/or spectroscopy tools play indispensable roles in nanoscience and nanotechnology research, as they provide rich information about material processes and properties. However, the interpretation of imaging data heavily relies on the "intuition" of experienced researchers. As a result, many of the deep graphical features obtained through these tools are often unused because of difficulties in processing the data and finding the correlations. Such challenges can be well addressed by deep learning. In this work, the optical characterization of 2D materials is used as a case study, and a neural-network-based algorithm is demonstrated for the material and thickness identification of 2D materials with high prediction accuracy and real-time processing capability. Further analysis shows that the trained network can extract deep graphical features such as contrast, color, edges, shapes, flake sizes, and their distributions, based on which an ensemble approach is developed to predict the most relevant physical properties of 2D materials. Finally, a transfer learning technique is applied to adapt the pretrained network to other optical identification applications. This artificial-intelligence-based material characterization approach is a powerful tool that would speed up the preparation, initial characterization of 2D materials and other nanomaterials, and potentially accelerate new material discoveries.

Research on 2D materials has grown exponentially over the past decade.[1,2] Hundreds of 2D materials have been isolated and studied,[2,3] offering a wide range of optical and electronics properties, including metals, semiconductors, insulators, magnets, and superconductors. The most widely used approach to obtain high-quality 2D crystals today is mechanical exfoliation,[4] followed by 2D crystal "hunting" under an optical microscope (**Figure 1**, left). This task is time-consuming and difficult especially for inexperienced researchers, but tedious for experienced researchers. There has, therefore, been growing interest in automating such a process.[5] An automatic optical identification and/or characterization system requires an algorithm that performs reliably for different materials with high accuracy, is fast enough for real-time processing, and is easily adaptable to different optical setups and different user

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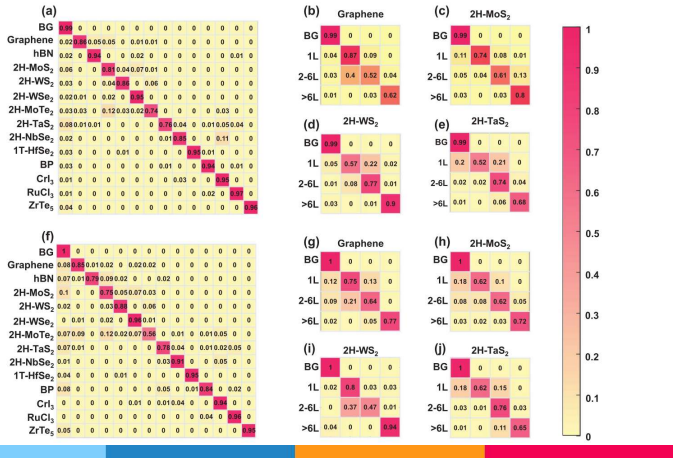
Deep-Learning-Enabled Fast Optical Identification and Characterization of 2D Materials



Adv. Mater. 2020, 2000953

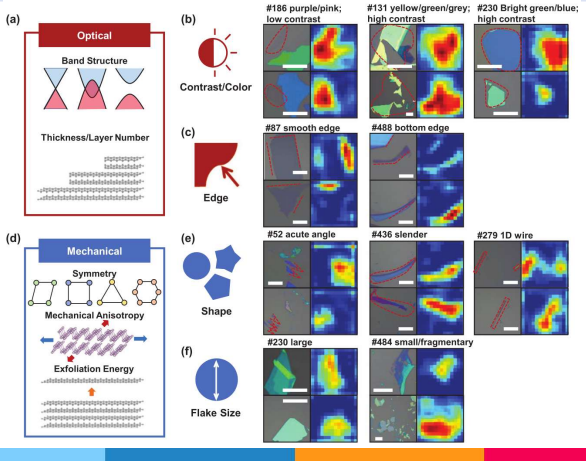
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Deep-Learning-Enabled Fast Optical Identification and Characterization of 2D Materials



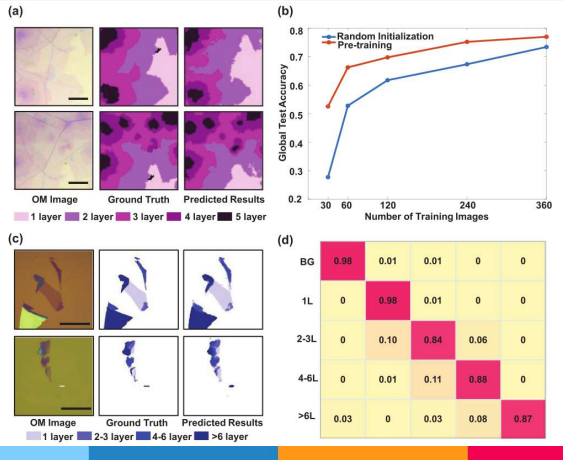
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Deep-Learning-Enabled Fast Optical Identification and Characterization of 2D Materials



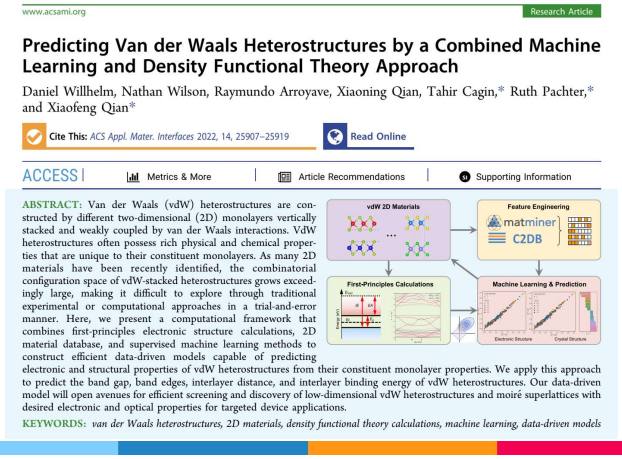
Deep-Learning-Enabled Fast Optical Identification and Characterization of 2D Materials

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研究案例⑤:机器学习范德华异质结

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研究案例⑤:机器学习范德华异质结

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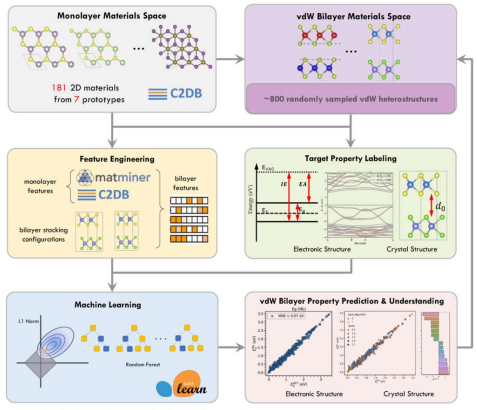


Figure 1. General schematic of computational and machine learning workflow.

研究案例⑤:机器学习范德华异质结

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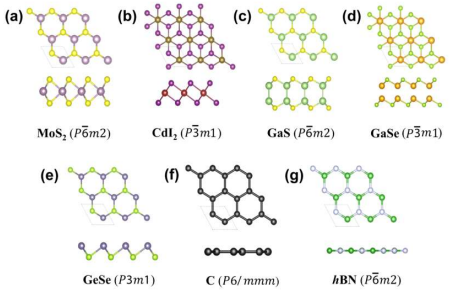


Figure 2. Crystal structures and symmetries of the monolayer prototypes used to create vdW heterostructures. (a) 1H-MoS_2, (b) 1T-CdI_2, (c) hexagonal GaSe, (d) hexagonal GaSe, (e) γ-GeSe, (f) graphene, and (g) *h*BN.

研究案例⑤:机器学习范德华异质结

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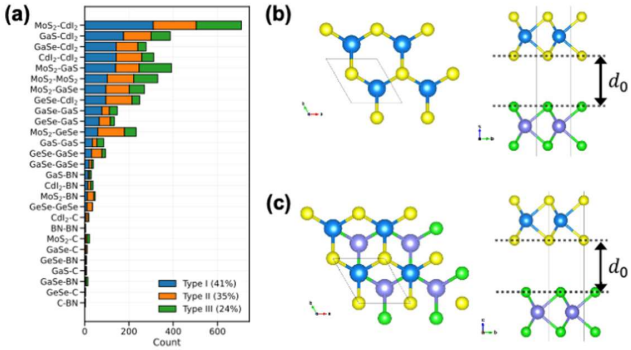


Figure 3. (a) Distribution of bilayer vdW heterostructures with Type I, II, and III band alignment. An illustration of top and side views of an example vdW heterostructure bilayer in the (b) AA stacking configuration and (c) AB stacking configuration. Interlayer distance d_0 is calculated as the vertical distance between the lowest atom of the top layer and the highest atom of the bottom layer.

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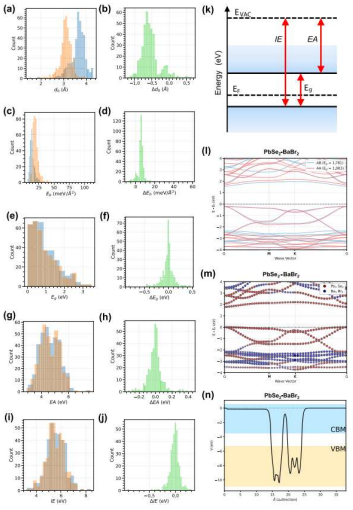


Figure 4. Distribution of the (a, b) interlayer distance d_0 and its difference between AA and AB ($\Delta d = d_0^{AA} - d_0^{AB}$), (c, d) interlayer binding energy E_b and its difference between AA and AB ($\Delta E_b = E_b^{AA} - E_b^{AB}$), (e, f) band gap E_g and its difference between AA and AB ($\Delta E_g = E_g^{AA} - E_g^{AB}$), (g, h) ionization energy IE and its difference between AA and AB ($\Delta IE = IE^{AA} - IE^{AB}$), and (i, j) electron affinity EA and its difference between AA and AB ($\Delta EA = EA^{AA} - EA^{AB}$). (k) Definitions of E_g, EA, and IE with respect to the vacuum level. From the high-throughput DFT, we collect the (l) AA vs AB band structure plot, (m) project band structure plot with respect to the constituent monolayers, and (n) local potential plot for each vdW heterostructure.

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研究案例⑤：机器学习范德华异质结

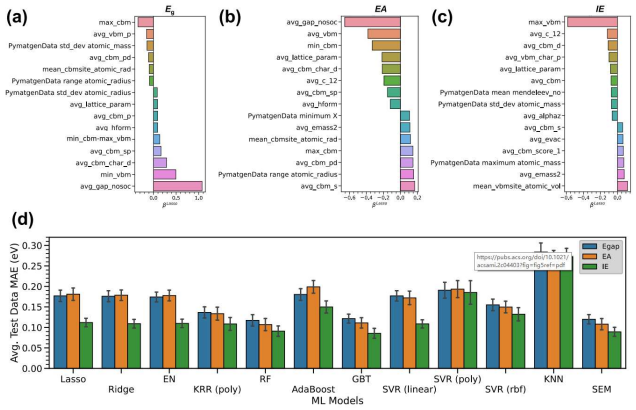


Figure 5. (a–c) Top 15 descriptors for the E_g, EA, and IE and their LASSO coefficients. (d) Average repeated k-fold cross-validation MAE for different machine learning algorithms. The black vertical line represents the standard deviation of the testing MAE.

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研究案例⑤：机器学习范德华异质结

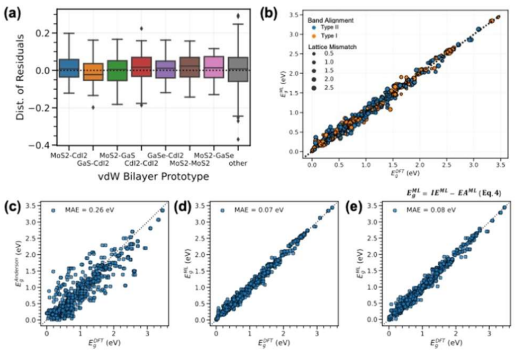
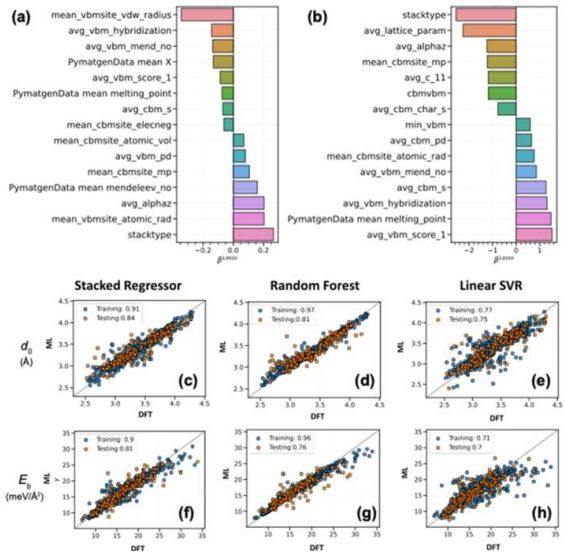


Figure 6. (a) Distribution of band gap residuals for different bilayer prototypes. (b) DFT- and ML-predicted band gap where the orange and blue colors represent Type I and Type II band alignments, respectively. The size of the datapoint corresponds to the lattice mismatch percent. The prediction of E_g using (c) Anderson's rule, (d) machine learning model of E_g, and (e) machine learning model of EA and IE with E_g defined by E_g = EA − IE.

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研究案例⑤：机器学习范德华异质结

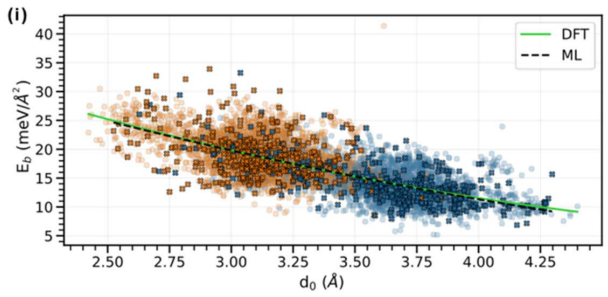


Figure 7. Top 15 descriptors for the (a) d_0 and (b) E_b models via the LASSO estimator descriptor selection. Comparison of the DFT and ML predictions using the Stacked Regressor, RF, and linear SVR models for the (c–e) d_0 and (f–h) E_b target properties. (i) Binding energy (E_b) vs the interlayer distance (d_0). The "x" data points correspond to the training data, while "o" represents the predictions on the unlabeled bilayers. Orange and blue colors represent AB and AA stacked bilayer vdW heterostructures, respectively.

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研究案例⑤：机器学习范德华异质结

Table 3. Average Testing (Training) MAE from Repeated k-Fold Cross-Validation for Interlayer Distance d_0 and Interlayer Binding Energy E_b Models

machine learning algorithm	d_0 MAE (Å)	E_b MAE (meV/Å^2)
Stacked Ensemble Method (SEM)	0.119 (0.078)	1.387 (0.902)
Random Forest	0.119 (0.044)	1.514 (0.564)
Gradient Boosted Trees	0.122 (0.067)	1.496 (0.817)
Kernel Ridge (poly)	0.130 (0.078)	1.438 (0.826)
SVR (linear)	0.136 (0.122)	1.707 (1.534)
Elastic Net	0.139 (0.124)	1.784 (1.631)
Ridge	0.140 (0.125)	1.786 (1.638)
LASSO	0.141 (0.125)	1.807 (1.633)
Ordinary Least Squares	0.142 (0.125)	1.799 (1.633)
SVR (rbf)	0.152 (0.099)	1.974 (1.639)
AdaBoost	0.156 (0.133)	2.278 (1.980)
SVR (poly)	0.193 (0.121)	2.458 (1.971)
K-Neighbors Regressor	0.294 (0.237)	2.953 (2.392)

- ✓ 4,000 unique bilayer vdW heterostructures
- ✓ machine learning models predict the bilayer Eg, IE, EA, d0, and Eb with low error

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研究案例⑥：无机材料设计通用模型

r1-sci] 29 Jan 2024

MatterGen: a generative model for inorganic materials design

Claudio Zeni[1†], Robert Pinsler[1†], Daniel Zügner[1†], Andrew Fowler[1†], Matthew Horton[1†], Xiang Fu[1], Aliaksandra Shysheya[1], Jonathan Crabbé[1], Lixin Sun[1], Jake Smith[1], Bichlien Nguyen[1], Hannes Schulz[1], Sarah Lewis[1], Chin-Wei Huang[1], Ziheng Lu[1], Yichi Zhou[1], Han Yang[1], Hongxia Hao[1], Jielan Li[1], Ryota Tomioka[1*†], Tian Xie[1*†]

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研究案例⑥：无机材料设计通用模型

Article

A generative model for inorganic materials design

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Claudio Zeni[1,4], Robert Pinsler[1,4], Daniel Zügner[1,4], Andrew Fowler[1,4], Matthew Horton[1,4], Xiang Fu[1], Zilong Wang[2], Aliaksandra Shysheya[1], Jonathan Crabbé[1], Shoko Ueda[1], Roberto Sordillo[1], Lixin Sun[1], Jake Smith[2], Bichlien Nguyen[2], Hannes Schulz[2], Sarah Lewis[1], Chin-Wei Huang[2], Ziheng Lu[2], Yichi Zhou[2], Han Yang[2], Hongxia Hao[2], Jielan Li[2], Chunlei Yang[2], Wenjie Li[2], Ryota Tomioka[1,2] ✉ & Tian Xie[1,2] ✉

The design of functional materials with desired properties is essential in driving technological advances in areas such as energy storage, catalysis and carbon capture[1–3]. Generative models accelerate materials design by directly generating new materials given desired property constraints, but current methods have a low success rate in proposing stable crystals or can satisfy only a limited set of property constraints[4–11]. Here we present MatterGen, a model that generates stable, diverse inorganic materials across the periodic table and can further be fine-tuned to steer the generation towards a broad range of property constraints. Compared with previous generative models[4,12], structures produced by MatterGen are more than twice as likely to be new and stable, and more than ten times closer to the local energy minimum. After fine-tuning, MatterGen successfully generates stable, new materials with desired chemistry, symmetry and mechanical, electronic and magnetic properties. As a proof of concept, we synthesize one of the generated structures and measure its property value to be within 20% of our target. We believe that the quality of generated materials and the breadth of abilities of MatterGen represent an important advancement towards creating a foundational generative model for materials design.

https://www.nature.com/articles/s41586-025-08628-5

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研究案例⑥：无机材料设计通用模型

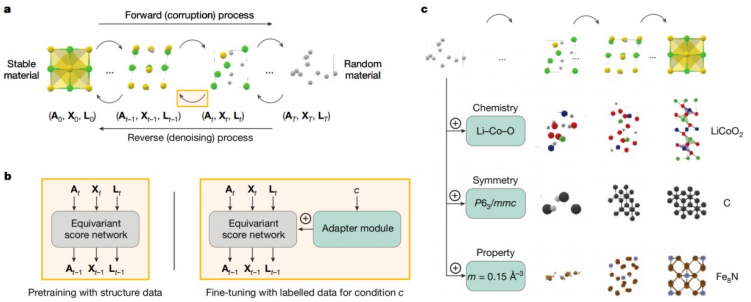
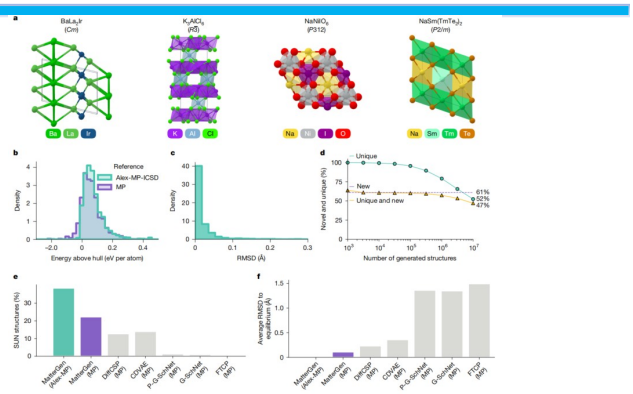


Fig. 1 | Inorganic materials design with MatterGen. a, MatterGen generates stable materials by reversing a corruption process through iteratively denoising a random structure. The forward diffusion process independently corrupts atom types **A**, coordinates **X** and the lattice **L** towards a physically motivated distribution of random materials. **b**, An equivariant score network is pretrained on a large dataset of stable material structures to jointly denoise atom types, coordinates and the lattice. The score network is then fine-tuned with a labelled dataset through an adapter module that adapts the model using the encoded property *c*. **c**, The fine-tuned model generates materials with desired chemistry, symmetry or scalar property constraints. *m*, magnetic density.

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研究案例⑥：无机材料设计通用模型



stable,
unique,
novel,
(S.U.N.)
materials

Fig. 2 | Generating stable, unique and new inorganic materials. **a**, Visualization of four randomly selected crystals generated by MatterGen, with corresponding reduced formula and space group. **b**, Distribution of energy above hull values of generated structures using MP and Alex-MP-ICSD datasets as energy references, respectively. **c**, Distribution of root mean squared displacement (RMSD) between initial generated and DFT-relaxed structures. **d**, Percentage of unique, new structures as a function of the number of generated structures. **e**, **f**, Percentage of SUN structures (**e**) and average RMSD between initial and DFT-relaxed structures (**f**) for MatterGen, MatterGen-MP and several baseline models, including DiffCSP[35], CDVAE[4], P-G-SchNet, G-SchNet[36] and FTCP[37]. Training datasets are in parentheses. Percentage of SUN structures are computed using 1,024 samples for MatterGen and 1,000 for baseline models.

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研究案例⑥：无机材料设计通用模型

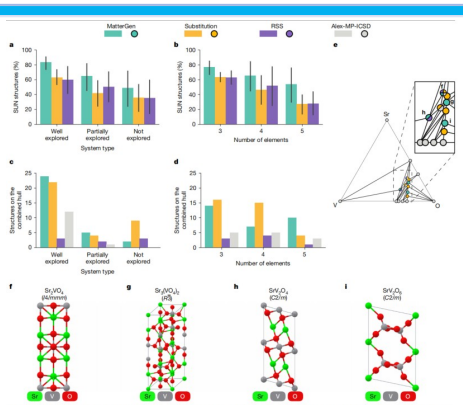


Fig. 3 | Generating materials in target chemical system. a, **b**, Mean percentage of SUN structures generated by MatterGen and baselines for 27 chemical systems, grouped by system type (**a**) and number of elements (**b**). Percentages are computed on 100 structures for each of 9 chemical systems. Error bars denote 95% percentile intervals ($n = 9$). **c**, **d**, Number of structures on the combined convex hull found by each method and in the Alex-MP-ICSD dataset, grouped by system type (**c**) and number of elements (**d**). **e**, Convex hull diagram for V–Sr–O, a well-explored ternary system. Dots denote structures on the hull, their coordinates show the element ratio of their composition and their colour indicates by which method they were discovered. **f**–**i**, Four structures that MatterGen discovered (rediscovered in the case of **f**) on the V–Sr–O hull shown in **e**, along with their reduced formula: Sr_2VO_4 (**f**), $Sr_3(VO_4)_2$ (**g**), SrV_2O_4 (**h**) and SrV_4O_8 (**i**).

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研究案例⑥：无机材料设计通用模型

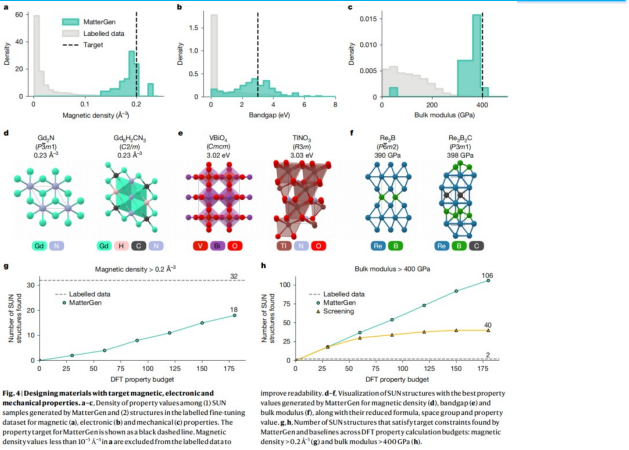


Fig. 4 | Designing materials with target magnetic, electronic and mechanical properties. a–**c**, Density of property values among (1) SUN samples generated by MatterGen and (2) structures in the labelled fine-tuning dataset for magnetic (**a**), electronic (**b**) and mechanical (**c**) properties. The property target for MatterGen is shown as a black dashed line. Magnetic density values less than 10^{-2} Å^{-3} in **a** are excluded from the labelled data to improve readability. **d**–**f**, Visualization of SUN structures with the best property values generated by MatterGen for magnetic density (**d**), bandgap (**e**) and bulk modulus (**f**), along with their reduced formula, space group and property value. **g**, **h**, Number of SUN structures that satisfy target constraints found by MatterGen and baselines across DFT property calculation budgets: magnetic density > 0.2 Å^{-3} (**g**) and bulk modulus > 400 GPa (**h**).

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研究案例⑥：无机材料设计通用模型

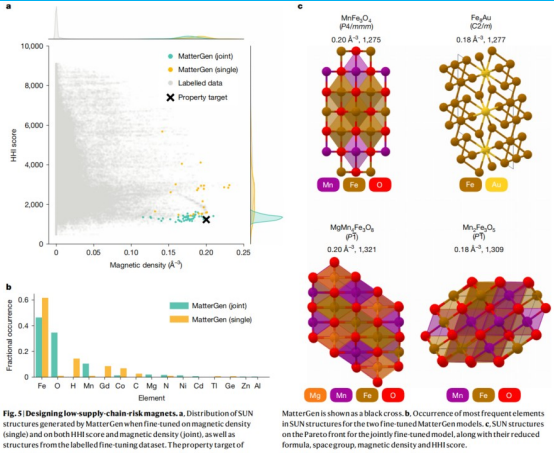


Fig. 5 | Designing low-supply-chain-risk magnets. a, Distribution of SUN structures generated by MatterGen when fine-tuned on magnetic density (single) and on both HHI score and magnetic density (joint), as well as structures from the labelled fine-tuning dataset. The property target of MatterGen is shown as a black cross. **b**, Occurrence of most frequent elements in SUN structures for the two fine-tuned MatterGen models. **c**, SUN structures on the Pareto front for the jointly fine-tuned model, along with their reduced formula, space group, magnetic density and HHI score.

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研究案例⑥：无机材料设计通用模型

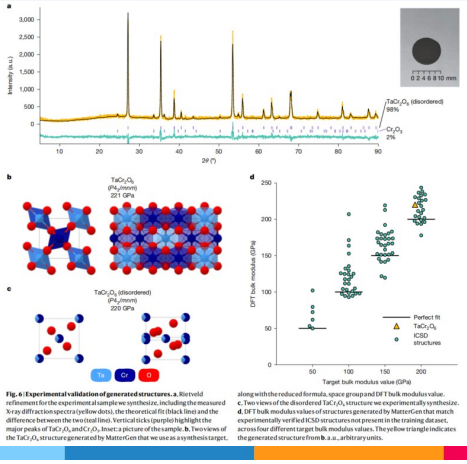


Fig. 6 | Experimental validation of generated structures. a, Rietveld refinement for the experimental sample we synthesize, including the measured X-ray diffraction spectra (yellow dots), the theoretical fit (black line) and the difference between the two (teal line). Vertical ticks (purple) highlight the major peaks of $TaCr_2O_6$ and Cr_2O_3. Inset: a picture of the sample. b, Two views of the $TaCr_2O_6$ structure generated by MatterGen that we use as a synthesis target, along with the reduced formula, space group and DFT bulk modulus value. c, Two views of the disordered $TaCr_2O_6$ structure we experimentally synthesize. d, DFT bulk modulus values of structures generated by MatterGen that match experimentally verified ICSD structures not present in the training dataset, across four different target bulk modulus values. The yellow triangle indicates the generated structure from b. a.u., arbitrary units.

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研究案例⑦：深度学习发现稳定晶体

Article

Scaling deep learning for materials discovery

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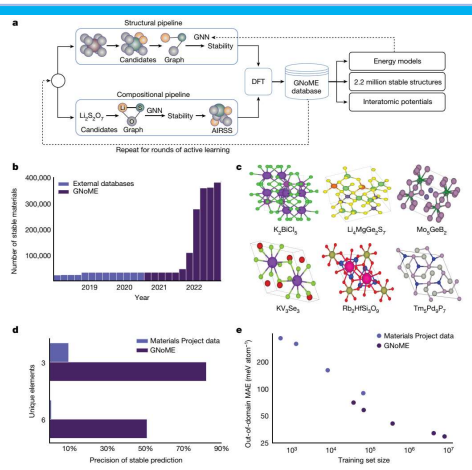
Amil Merchant[1,3✉], Simon Batzner[1,3], Samuel S. Schoenholz[1,3], Muratahan Aykol[1], Gowoon Cheon[2] & Ekin Dogus Cubuk[1,3✉]

Novel functional materials enable fundamental breakthroughs across technological applications from clean energy to information processing[1–11]. From microchips to batteries and photovoltaics, discovery of inorganic crystals has been bottlenecked by expensive trial-and-error approaches. Concurrently, deep-learning models for language, vision and biology have showcased emergent predictive capabilities with increasing data and computation[12–14]. Here we show that graph networks trained at scale can reach unprecedented levels of generalization, improving the efficiency of materials discovery by an order of magnitude. Building on 48,000 stable crystals identified in continuing studies[11,15–17], improved efficiency enables the discovery of 2.2 million structures below the current convex hull, many of which escaped previous human chemical intuition. Our work represents an order-of-magnitude expansion in stable materials known to humanity. Stable discoveries that are on the final convex hull will be made available to screen for technological applications, as we demonstrate for layered materials and solid-electrolyte candidates. Of the stable structures, 736 have already been independently experimentally realized. The scale and diversity of hundreds of millions of first-principles calculations also unlock modelling capabilities for downstream applications, leading in particular to highly accurate and robust learned interatomic potentials that can be used in condensed-phase molecular-dynamics simulations and high-fidelity zero-shot prediction of ionic conductivity.

[1]Google DeepMind, Mountain View, CA, USA. [2]Google Research, Mountain View, CA, USA. [3]These authors contributed equally: Amil Merchant, Simon Batzner, Samuel S. Schoenholz, Ekin Dogus Cubuk. ✉e-mail: amilmerchant@google.com; cubuk@google.com

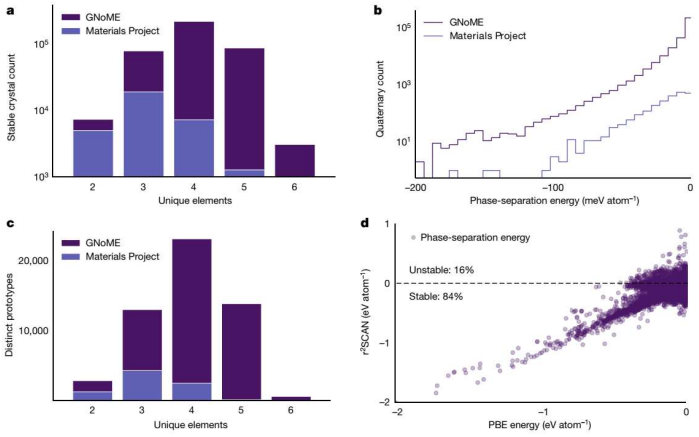
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研究案例⑦：深度学习发现稳定晶体



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研究案例⑦：深度学习发现稳定晶体



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研究案例⑦：深度学习发现稳定晶体

Mater. Futures 3 (2024) 027501 Commentary

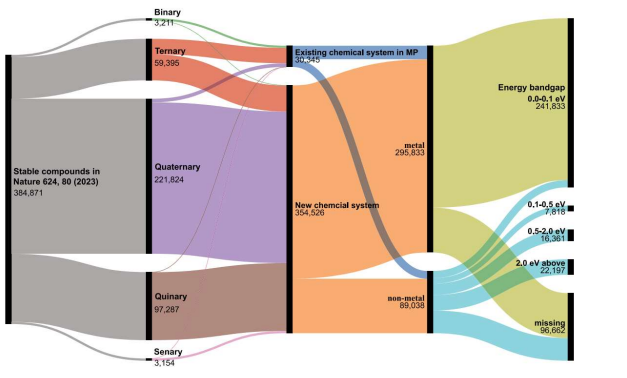


Figure 1. A close look at the GNoME dataset on its chemical system, metallicity, and energy band gap.

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研究案例⑧：机器学习有机发光材料

ARTICLES
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nature materials

Design of efficient molecular organic light-emitting diodes by a high-throughput virtual screening and experimental approach

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Virtual screening is becoming a ground-breaking tool for molecular discovery due to the exponential growth of available computer time and constant improvement of simulation and machine learning techniques. We report an integrated organic functional material design process that incorporates theoretical insight, quantum chemistry, cheminformatics, machine learning, industrial expertise, organic synthesis, molecular characterization, device fabrication and optoelectronic testing. After exploring a search space of 1.6 million molecules and screening over 400,000 of them using time-dependent density functional theory, we identified thousands of promising novel organic light-emitting diode molecules across the visible spectrum. Our team collaboratively selected the best candidates from this set. The experimentally determined external quantum efficiencies for these synthesized candidates were as large as 22%.

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研究案例⑧:机器学习有机发光材料

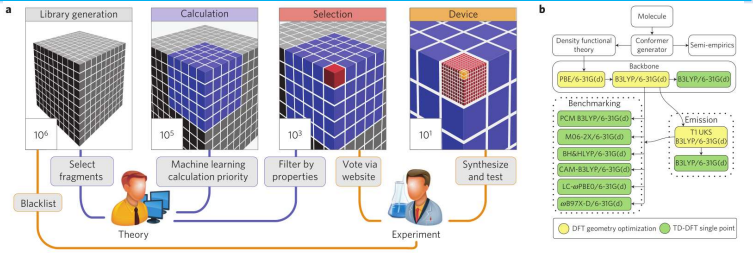


Figure 1 | Discovery pipeline. a, Diagram of the collaborative discovery approach: the search space decreases by over five orders of magnitude as the screening progresses. The cubes represent the size of the chemical space considered at any given stage of the process. The distinct screening stages, from left to right, involve different theoretical and computational approaches as well as experimental input and testing. b, Dependency tree for the quantum chemistry calculations employed in this study. The calculations labeled as backbone were performed for all analyzed molecules, leading compounds were also characterized using the methods labeled emission, and the benchmarking calculations were used to assess predictive power.

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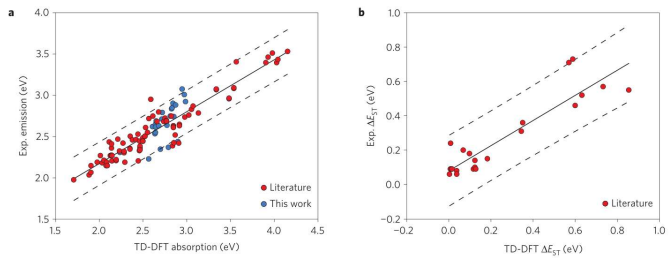


Figure 2 | Experiment–theory calibration. a, TD-DFT/B3LYP/6-31G(d) vertical absorption against photoluminescent emission maximum in toluene solution. Data in red represent measurements reported in the literature, while data in blue correspond to compounds synthesized in this screening work. The lines indicate a linear fit against the literature data (solid) with 95% confidence bounds (dashed). **b**, Experiment–theory comparison of TD-DFT/B3LYP/6-31G(d) singlet–triplet gap against experimental values, determined via the thermal activation energy method, measured in frozen toluene solution. The lines indicate a linear fit against the literature data (solid) with 95% confidence bounds (dashed).

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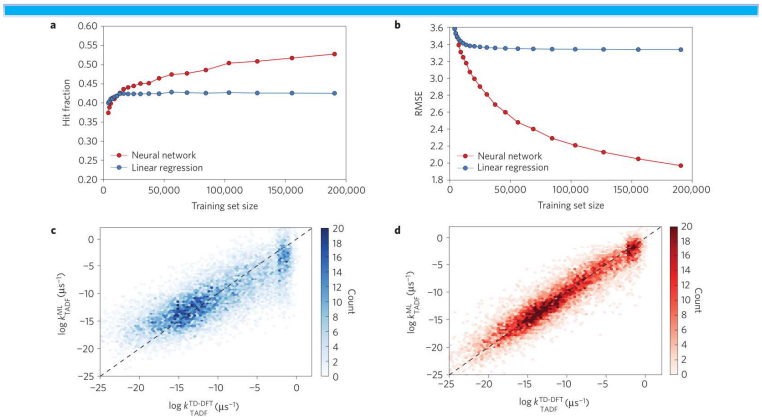


Figure 3 | Effectiveness of machine learning. a, Fraction of molecules in the test set correctly ranked in the top 5%, as a function of the amount of training data. **b**, Root mean square error (RMSE) in $\log(k_{TADF})$ as a function of the amount of training data. **c**, Linear model predictions against the TD-DFT-derived data with the largest training set. $R^2 = 0.80$. **d**, Neural network predictions against the TD-DFT-derived data with the largest training set. $R^2 = 0.94$.

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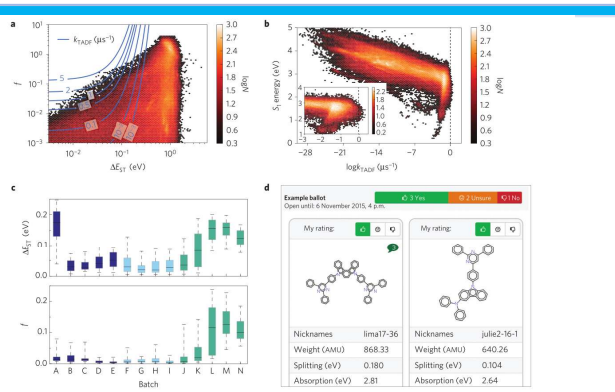


Figure 4 | Candidate statistics and voting tool. a, Number of screened molecules as a function of singlet–triplet splitting (ΔE_{ST}) and oscillator strength (f). Contour lines represent estimated k_{TADF} (μs^{-1}) assuming S_1 at 3.0 eV. **b**, Number of screened molecules as a function of k_{TADF} and S_1 energy. Vertical dashed line corresponds to $k_{TADF} = 1 \mu s^{-1}$. **c**, ΔE_{ST} and f for decision batches A–N, colour-coded by each of the generations in the candidate library. The box-and-whisker plots represent the statistical distribution of predictions for all the candidates in each batch. The bottom and top of the bar are the first and third quartiles, and the band inside the box is the median. The lines extending vertically from the boxes indicate the maximum and minimum of the range. **d**, Screenshot from the interactive web tool for molecular voting.

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Table 1 | Selected molecules and devices tested.

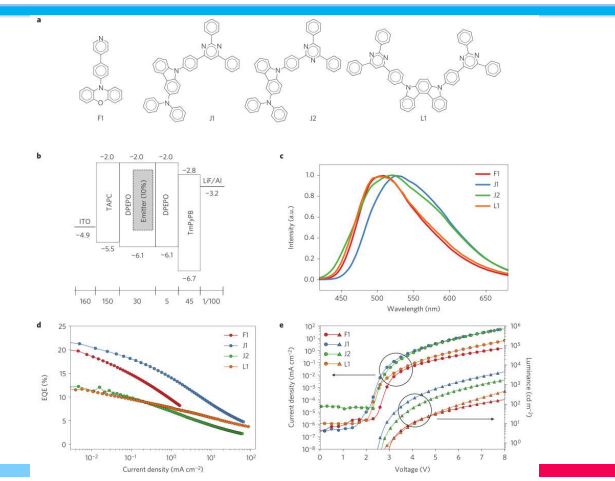
	Theory				Experiment										
					ΔE_{ST} Tol. (meV)		PL_{max} (nm)*		PLQY Tol. (%)		f				
Cpd	PL (nm)	ΔE_{ST} (meV)	f	k_{TADF} (μs^{-1})	Onset	Max	Tol.	Thin film	Air	N_2	Abs†	Em‡	τ_p (ns)	τ_d (µs) [k_d]	EL_{max} (nm)§	Max EQE
F1	485	15	0.003	0.141	100	−160	486	456‖	10	13	0.020	0.016	6.6	6.1 [0.16]	507	20.6
J1	485	104	0.124	0.221	60	−140	490	503‖	42	67	0.153	0.060	7.4	2.2 [0.45]	524	22.0
J2	485	36	0.001	0.004	180	0	462	502‖	24	40	0.051	0.025	10	9.2 [0.11]	520	12.7
L1	451	179	0.257	0.029	320	110	451	494¶	48	67	0.107	0.040	11	9.5 [0.10]	501	11.9

*Maximum wavelength of photoluminescence. †Determined from absorption spectra in toluene solution (details in section 1.2 of the Supplementary Information). ‡Determined from prompt rate in DPEPO films (details in section 1.2 of the Supplementary Information). §Maximum wavelength of electroluminescence. ‖Drop-cast thin film. ¶Co-evaporation in DPEPO. $\tau_p = \tau_{prompt}$; $\tau_d = \tau_{delay}$; $k_d = k_{delay}$.

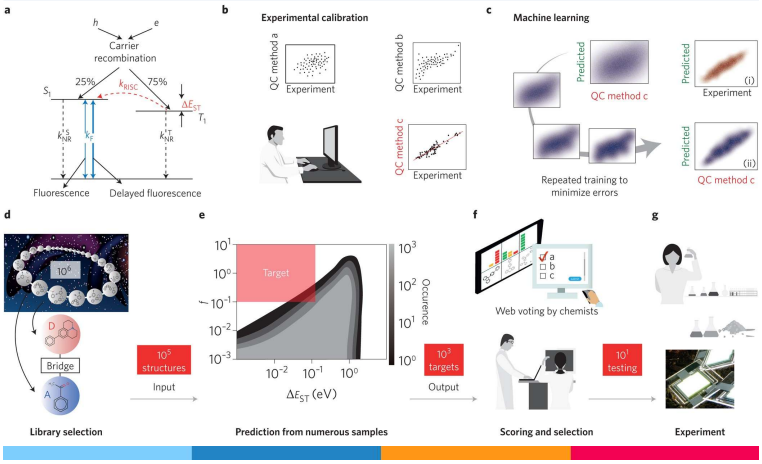
An integrated computation-driven search targeting novel TADF OLED emitters. The discovery process has explored over **a million candidates** by combining hundreds of thousands of quantum simulations, machine learning, and domain expertise from synthetic chemists, device scientists and industry partners, and ultimately the experimental realization of lead candidates. Excellent predictive power has resulted in the report of devices with **over 22% EQE**.

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Summary-2

- New 2D materials from data mining / machine learning
- High Curie-temperature magnetic materials
- AI-assisted Materials identification and characterization
- Other new materials with exceptional properties: Superconductors, optical/electric/structural materials?

总结

- 基于已有实验合成材料数据库的高通量筛选
- 基于理论预测二维材料数据库的数据挖掘和机器学习
- 高通量计算、大数据技术/机器学习可加速新材料研发

Nature 2025, 639, 624-632; *arXiv*:2312.03687; *Nature* 2023, 624 80–85.
ACS Appl. Mater. Interfaces 2022, 14, 25907−25919.

- 基于机器学习的材料表征/设计/验证

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谢谢大家!
Thank you for your attention!

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